CLINICAL CONDITIONS

Gastrointestinal Practice Tests

3 tests, 238 USMLE-style questions, with answers and explanations

For educational use only. This is a study aid, not a clinical reference.

Instructions

120 minutes
for each test

79–80 questions
in each test

About 90 seconds
for each question

Taking a test

- Take each test in one sitting, with a timer set to 120 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Mark your answers on the answer sheet after the last question of each test, or on a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

About the tests

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test covers all 28 conditions, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- Questions are in random order, and the condition being tested is not shown until the answers.

Scoring and review

- Your score is the number of correct answers. Use the answer key at the start of each answer section, and write your score on your answer sheet.
- Then read the explanations. Each one names the condition and step, so you can see which conditions to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

Contents

Test A

Test B

Test C

Test A answers

Test B answers

Test C answers

Abbreviations

You may refer to this page during a test.

5-ASA 5-aminosalicylic acid

5-HIAA 5-hydroxyindoleacetic acid

A1AT Alpha-1 antitrypsin

ABX Antibiotics

AFP Alpha-fetoprotein

ALP Alkaline phosphatase

ALT Alanine aminotransferase

AMA Anti-mitochondrial antibody

ANA Antinuclear antibody

anti-CCP Anti-cyclic citrullinated peptide antibody

anti-DGP Anti-deamidated gliadin peptide antibody

anti-dsDNA Anti-double-stranded DNA antibody

anti-HAV Antibody to hepatitis A virus

anti-HBc Antibody to hepatitis B core antigen

anti-HBe Antibody to hepatitis B e antigen

anti-HBs Antibody to hepatitis B surface antigen

Anti-HCV Antibody to hepatitis C virus

anti-HDV Antibody to hepatitis D virus

anti-tTG Anti-tissue transglutaminase antibody

AST Aspartate aminotransferase

BMI Body mass index

BUN Blood urea nitrogen

CA 19-9 Cancer antigen 19-9

cAMP Cyclic adenosine monophosphate

cccDNA Covalently closed circular DNA: a stable template that persists in the nucleus

CD4 Cluster of differentiation 4

CN X Cranial nerve X (the vagus nerve)

COPD Chronic obstructive pulmonary disease

CREST Calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia

CRP C-reactive protein

CT Computed tomography

CT AP Computed tomography of the abdomen and pelvis

CYP2E1 Cytochrome P450 2E1

DNA Deoxyribonucleic acid

DVT Deep vein thrombosis

ECG Electrocardiogram

ERCP Endoscopic retrograde cholangiopancreatography

FAST Focused assessment with sonography in trauma

FBC Full blood count

FIB-4 Fibrosis-4 index

FODMAP Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols

GE Gastroesophageal

GERD Gastroesophageal reflux disease

GGT Gamma-glutamyl transferase

GI Gastrointestinal

HBeAg Hepatitis B e antigen: active replication, high infectivity

HBIG Hepatitis B immune globulin

HBsAg Hepatitis B surface antigen: current infection

HBV Hepatitis B virus

HBV DNA Viral load: active replication

HCV Hepatitis C virus

HDV Hepatitis D (delta) virus

HELLP Hemolysis, elevated liver enzymes, low platelets

HIDA Hepatobiliary iminodiacetic acid scan (cholescintigraphy)

HIV Human immunodeficiency virus

HLA Human leukocyte antigen

IBD Inflammatory bowel disease

IBS Irritable bowel syndrome

ICU Intensive care unit

IgA Immunoglobulin A

IgG Immunoglobulin G

IgG4 Immunoglobulin G4

IgM Immunoglobulin M

INR International normalized ratio

IV Intravenous

JAK2 Janus kinase 2 (V617F mutation)

JAK2 V617F Janus kinase 2 V617F mutation, found in most myeloproliferative neoplasms

LCHAD Long-chain 3-hydroxyacyl-CoA dehydrogenase, an enzyme of fatty acid beta-oxidation

LDH Lactate dehydrogenase

LDL Low-density lipoprotein

LES Lower esophageal sphincter

LFTs Liver function tests

LLQ Left lower quadrant

MALT Mucosa-associated lymphoid tissue

MASLD Metabolic dysfunction-associated steatotic liver disease (formerly NAFLD)

MCAD Medium-chain acyl-CoA dehydrogenase

MCV Mean corpuscular volume

MRCP Magnetic resonance cholangiopancreatography

MRI Magnetic resonance imaging

NAD Nicotinamide adenine dinucleotide

NADH Reduced nicotinamide adenine dinucleotide

NAPQI N-acetyl-p-benzoquinone imine

NPO Nil per os: nothing by mouth

NSAID Nonsteroidal anti-inflammatory drug

NSAIDs Nonsteroidal anti-inflammatory drugs

NTCP Sodium taurocholate co-transporting polypeptide, a bile salt transporter on hepatocytes

p-ANCA Perinuclear anti-neutrophil cytoplasmic antibody

PAS Periodic acid-Schiff

PCR Polymerase chain reaction

PET Positron emission tomography

PPI Proton pump inhibitor

PT Prothrombin time

PTT Partial thromboplastin time

R-CHOP Rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone

RLQ Right lower quadrant

RNA Ribonucleic acid

RUQ Right upper quadrant

SMA Superior mesenteric artery

TIPS Transjugular intrahepatic portosystemic shunt

TPMT Thiopurine methyltransferase

TPN Total parenteral nutrition

VIP Vasoactive intestinal peptide

WBC White blood cells

Test A

80 questions · 120 minutes

1. A 27-year-old woman has 8 months of recurrent lower abdominal cramping that improves after bowel movements, with alternating loose and hard stools. She has no weight loss, bleeding, or nocturnal symptoms. FBC, CRP, fecal calprotectin, and celiac serology are normal.

Which is the most appropriate next step?

A. CT AP
B. Empiric metronidazole for small bowel overgrowth
C. Start oral prednisone for presumed IBD
D. Colonoscopy
E. Reassurance and a trial of a low-FODMAP diet

2. A 58-year-old man has had untreated achalasia for 25 years, with a markedly dilated esophagus full of retained food.

He is at increased risk of which malignancy?

A. Leiomyosarcoma of the esophageal muscle wall
B. Hepatocellular carcinoma
C. Adenocarcinoma of the proximal stomach
D. Gastric MALT lymphoma
E. Squamous cell carcinoma of the esophagus

3. A 47-year-old man who drinks a bottle of vodka daily has fever, jaundice, and tender hepatomegaly after a week of heavy drinking. AST is 240 U/L, ALT is 95 U/L, GGT is elevated, and MCV is 104 fL.

Which is the most likely diagnosis?

A. Paracetamol toxicity
B. Ischemic hepatitis
C. Alcoholic hepatitis
D. Acute viral hepatitis A
E. MASLD

4. A 52-year-old woman with Hashimoto thyroiditis has 6 months of fatigue and generalized itching. She has xanthelasma around her eyes. ALP is 610 U/L, GGT is elevated, and AST/ALT are mildly elevated. Right upper quadrant ultrasound shows no bile duct dilation.

Which antibody is most likely to be positive?

A. Anti-liver/kidney microsomal antibody type 1
B. Anti-smooth muscle antibody
C. Anti-tissue transglutaminase
D. Anti-mitochondrial antibody
E. p-ANCA

5. During a cruise, 120 passengers develop sudden vomiting and watery diarrhea within 1–2 days of each other. Symptoms resolve within 72 hours.

Which organism is the most likely cause?

A. Entamoeba histolytica
B. Clostridioides difficile
C. Norovirus
D. Rotavirus
E. Shiga toxin-producing E. coli

6. A patient with cirrhosis has bilirubin 2.5 mg/dL, albumin 3.0 g/dL, INR 1.5, mild ascites controlled with diuretics, and no encephalopathy.

Which Child-Pugh class is this?

A. Class A
B. The score cannot be calculated without creatinine
C. Class C
D. Class D
E. Class B

7. A 26-year-old woman in her first pregnancy, at 35 weeks, has 3 days of nausea, vomiting, and malaise, and is now jaundiced and confused. Glucose is 2.4 mmol/L (43 mg/dL), INR is 2.1, AST is 480 U/L, platelets are 160000/µL and blood pressure is 134/86 mm Hg.

Which is the most appropriate definitive management?

A. N-acetylcysteine and observation
B. IV magnesium sulfate and expectant management
C. Ursodeoxycholic acid
D. Corticosteroids for fetal lung maturity, then deliver at 37 weeks
E. Prompt delivery

8. A 66-year-old man has diverticulosis on a screening colonoscopy. He has occasional constipation but no history of diverticulitis or bleeding.

Which recommendation is most appropriate?

A. Increase dietary fiber
B. Avoid all nuts and seeds
C. Long-term prophylactic ABX
D. Elective sigmoid colectomy
E. Repeat colonoscopy every year

9. A 38-year-old man with untreated HIV (CD4 count 45/μL) has months of severe, watery diarrhea and weight loss. A modified acid-fast stain of stool shows small, round oocysts.

Which organism is the most likely cause?

A. Cryptosporidium
B. Entamoeba histolytica
C. Strongyloides stercoralis
D. Giardia lamblia
E. Mycobacterium tuberculosis

10. A 72-year-old man with a lifelong low-fiber diet and chronic constipation is found to have many outpouchings in the sigmoid colon on colonoscopy.

Which statement best describes these outpouchings?

A. False diverticula of mucosa and submucosa herniating at vasa recta entry points
B. True diverticula containing all layers of the bowel wall, including the muscularis externa
C. Full-thickness herniation through the linea alba
D. Outpouchings of the posterior pharynx above the cricopharyngeus
E. Congenital remnants of the vitelline duct

11. A 70-year-old man taking an over-the-counter antacid several times a day for dyspepsia develops troublesome diarrhea. His previous antacid had caused constipation.

Which antacid is he most likely taking now?

A. Bismuth subsalicylate
B. Aluminum hydroxide
C. Magnesium hydroxide
D. Sucralfate
E. Calcium carbonate

12. A 40-year-old woman receiving chemotherapy for lymphoma develops severe odynophagia. Endoscopy shows multiple small, sharply demarcated "punched-out" ulcers in the mid esophagus. Biopsy from the ulcer edge shows multinucleated giant cells with intranuclear inclusions.

Which organism is the most likely cause?

A. Herpes simplex virus
B. Helicobacter pylori
C. Candida albicans
D. Epstein-Barr virus
E. Cytomegalovirus

13. A 66-year-old woman has 2 days of left lower quadrant pain, fever of 38.4°C and leukocytosis. She has tenderness in the LLQ without rebound or guarding.

Which is the most appropriate diagnostic test?

A. Flexible sigmoidoscopy
B. Abdominal ultrasound
C. CT AP with contrast
D. Colonoscopy
E. Barium enema

14. A 34-year-old man with Crohn disease, who had 60 cm of terminal ileum resected, presents with colicky flank pain and hematuria. Imaging shows a radiopaque kidney stone.

Which type of stone is most likely?

A. Calcium phosphate
B. Cystine
C. Struvite
D. Calcium oxalate
E. Uric acid

15. A 51-year-old woman with obesity (BMI 36), type 2 diabetes and hypertriglyceridemia has an ALT of 68 U/L and AST of 45 U/L. She drinks alcohol rarely. Ultrasound shows a diffusely echogenic liver. Viral hepatitis serologies are negative.

Which is the most common cause of death in patients with this condition?

A. Hepatic encephalopathy
B. Cardiovascular disease
C. Hepatocellular carcinoma
D. Variceal hemorrhage
E. Acute liver failure

16. A 27-year-old woman is in the ICU after a traumatic brain injury with raised intracranial pressure. On day 2 she develops hematemesis, and endoscopy shows multiple gastric erosions.

Which mechanism most directly increases acid production in this patient?

A. Hypovolemia leading to mucosal ischemia
B. Increased vagal (CN X) stimulation of parietal cells
C. Decreased prostaglandin E2 synthesis
D. Increased secretin release
E. Autoantibodies against the H+/K+ ATPase of parietal cells

17. A 34-year-old woman has had pancolonic ulcerative colitis for 8 years. She feels well on mesalazine.

Which is the most appropriate cancer screening plan?

A. Annual fecal occult blood testing with colonoscopy if positive
B. No screening until symptoms develop
C. CT colonography every 5 years
D. Surveillance colonoscopy every 1–3 years starting now
E. Colonoscopy every 10 years starting at age 45

18. A 28-year-old woman who has taken oral contraceptives for 10 years presents with sudden right upper quadrant pain and hypotension. CT shows a 7-cm liver mass with surrounding hemorrhage.

Which is the most likely underlying lesion?

A. Hepatic adenoma
B. Angiosarcoma
C. Focal nodular hyperplasia
D. Hemangioma
E. Hepatocellular carcinoma

19. A 31-year-old man is brought in after a motorcycle collision. He has right upper quadrant tenderness and abrasions over the right lower ribs. Blood pressure is 128/76 mm Hg and heart rate 88/min, and they remain stable after 1 L of crystalloid.

Which is the most appropriate next step?

A. Upper endoscopy
B. Diagnostic peritoneal lavage
C. CT abdomen with IV contrast
D. Discharge with outpatient follow-up
E. Emergency laparotomy

20. A 68-year-old woman who smokes heavily and has peripheral artery disease has dull periumbilical pain starting 30 minutes after meals. She now avoids eating and has lost 9 kg over 6 months.

Which mechanism best explains her pain?

A. Peptic ulcer disease with pain triggered by post-meal acid secretion
B. Atherosclerotic mesenteric arteries that cannot meet post-meal oxygen demand
C. Gallstone obstructing the cystic duct when the gallbladder contracts after meals
D. Acute embolic occlusion of the SMA
E. Mesenteric venous thrombosis causing congestion of the bowel wall after meals

21. A 52-year-old man with known erosive esophagitis develops sudden severe chest pain after an episode of forceful vomiting. He is tachycardic and febrile, and crepitus is felt in the neck. Chest x-ray shows air in the mediastinum.

Which is the most appropriate test to confirm the suspected diagnosis?

A. 24-hour pH monitoring
B. Upper endoscopy with biopsy
C. Esophageal manometry
D. Water-soluble contrast esophagram
E. Barium swallow

22. A 45-year-old woman has 12 hours of RUQ pain and fever. Ultrasound shows gallstones but is otherwise inconclusive, so a HIDA scan is performed.

Which HIDA scan finding confirms acute cholecystitis?

A. Tracer does not reach the duodenum
B. Tracer is taken up by the spleen
C. Tracer leaks into the peritoneum
D. Tracer fills the gallbladder within 30 minutes
E. The gallbladder does not fill with tracer

23. A physiology lecture explains why the liver tolerates occlusion of the hepatic artery better than many other organs tolerate loss of their arterial supply.

Which vessel provides most of the liver's blood flow?

A. Inferior vena cava
B. Splenic artery
C. Hepatic artery proper
D. Portal vein
E. Hepatic veins

24. A 78-year-old woman with a history of gallstones has 3 days of colicky abdominal pain, distention, vomiting, and no passage of stool or gas. CT shows air in the biliary tree, dilated small bowel loops, and a 3-cm calcified stone in the right lower quadrant.

Where is the stone most likely lodged?

A. Duodenojejunal flexure
B. Pylorus
C. Sigmoid colon
D. Ampulla of Vater
E. Ileocecal valve

25. A 29-year-old woman is diagnosed with celiac disease on serology and duodenal biopsy. She asks how her condition will be monitored once she starts treatment.

Which is the best way to monitor her response to a gluten-free diet?

A. Rising HLA-DQ2 expression
B. Falling IgA anti-tTG levels
C. Repeat D-xylose test every month
D. Serial fecal elastase
E. Annual colonoscopy

26. A slit-lamp examination of a patient with Wilson disease shows a golden-brown ring at the edge of the cornea.

In which layer of the cornea is the copper deposited?

A. Lens capsule
B. Corneal stroma
C. Corneal epithelium
D. Descemet membrane
E. Bowman layer

27. A 72-year-old woman with long-standing pulmonary hypertension and severe tricuspid regurgitation has jugular venous distention, bilateral leg edema, and a tender, enlarged liver. Pressing on the liver increases her jugular venous pulsation. At autopsy, the cut surface of the liver shows dark red centrilobular areas alternating with pale tan areas.

Which mechanism best explains the liver findings?

A. Portal vein thrombosis
B. Fatty infiltration of hepatocytes
C. Thrombosis of the hepatic veins blocking outflow from the sinusoids
D. Raised central venous pressure transmitted to the hepatic veins
E. Occlusion of the hepatic artery

28. A 58-year-old man with cirrhosis has breast enlargement, small testes, reddened palms, and many small, spider-like blood vessels on his chest.

Which mechanism best explains these findings?

A. Decreased albumin synthesis by the failing liver
B. Vitamin K deficiency from reduced bile salt secretion
C. Increased estrogen from reduced hepatic clearance
D. Increased ammonia from reduced hepatic clearance
E. Portal hypertension

29. A 76-year-old woman with A-fib has acute mesenteric ischemia from an SMA embolus on CT angiography. She is tachycardic with rising lactate but no peritonitis yet.

Which is the most appropriate management?

A. Bowel rest and TPN only
B. Observation with serial lactate
C. Colonoscopy
D. Oral warfarin and outpatient follow-up
E. IV heparin and emergency revascularization

30. A 63-year-old woman who takes naproxen daily develops sudden, severe epigastric pain that spreads across the abdomen. She is tachycardic, and the abdomen is rigid with rebound tenderness. An upright chest x-ray shows air under the right hemidiaphragm.

In addition to IV fluids, IV PPI, and ABX, which is the most appropriate next step?

A. Barium swallow to locate the perforation
B. Emergency surgical repair
C. Urgent upper endoscopy
D. Nasogastric tube and serial abdominal examinations only
E. H. pylori testing and observation

31. On day 3 of admission for severe alcohol-related pancreatitis, a 52-year-old man becomes hypotensive. Examination shows bruising over both flanks and around the umbilicus. Hemoglobin has fallen from 14 to 9 g/dL.

Which best explains these findings?

A. Disseminated intravascular coagulation alone
B. Upper GI bleeding from a peptic ulcer
C. Retroperitoneal hemorrhage from necrotizing pancreatitis
D. Splenic rupture from trauma
E. Ruptured pancreatic pseudocyst into the stomach

32. A 29-year-old woman at 32 weeks' gestation reports new heartburn that is worse after meals and when lying flat.

Which combination of factors best explains her symptoms?

A. Eosinophilic infiltration of the esophagus
B. Decreased LES tone and increased intragastric pressure
C. Loss of myenteric inhibitory neurons
D. Increased LES tone and decreased intragastric pressure
E. Increased gastric acid secretion from H. pylori infection

33. A 34-year-old woman taking combined oral contraceptives has 5 days of worsening, poorly localized abdominal pain. She is heterozygous for factor V Leiden. CT shows thrombus in the superior mesenteric vein with bowel wall thickening.

Which mechanism best explains her bowel ischemia?

A. Venous congestion reducing oxygen delivery to the bowel
B. Arterial embolus from the left atrium blocking the SMA
C. Vasospasm from vasopressors
D. Atherosclerosis of the celiac trunk
E. Hypoperfusion from cardiogenic shock reducing bowel oxygen delivery

34. A 47-year-old woman has persistent regurgitation despite bd PPI therapy and weight loss. pH monitoring confirms pathologic acid reflux, and manometry shows normal peristalsis. She wishes to stop long-term medication.

Which of the following is the most appropriate treatment?

A. Botulinum toxin injection into the LES
B. Pneumatic dilation
C. Heller myotomy
D. Add a calcium channel blocker
E. Laparoscopic Nissen fundoplication

35. A 58-year-old man with celiac disease who has not followed a gluten-free diet presents with weight loss, abdominal pain, and a small bowel perforation. Pathology of the resected segment shows a T-cell malignancy.

Which is the most likely diagnosis?

A. Colorectal adenocarcinoma
B. Carcinoid tumor
C. Extranodal marginal zone (MALT) lymphoma
D. Enteropathy-associated T-cell lymphoma
E. Burkitt lymphoma

36. A 1-cm yellow nodule is found at the tip of the appendix on histology after a routine appendectomy for appendicitis in a 30-year-old woman. It is composed of nests of uniform cells that stain for chromogranin A.

Which is the most likely diagnosis?

A. Mucocele
B. Appendiceal neuroendocrine (carcinoid) tumor
C. Fecalith
D. Reactive lymphoid hyperplasia of the appendiceal wall
E. Appendiceal mucinous adenocarcinoma

37. A 33-year-old woman found to have hepatitis B during antenatal screening has these results: HBsAg positive, HBeAg positive, anti-HBe negative, IgG anti-HBc positive, HBV DNA very high. HBsAg has been positive for over a year.

Which best describes her infection?

A. Chronic infection with high replication and infectivity
B. Chronic infection with low replication and infectivity
C. Acute infection with active replication and high infectivity
D. Resolved infection
E. Window period

38. A 44-year-old man has recurrent peptic ulcers, including one in the proximal jejunum, along with chronic watery diarrhea. He is H. pylori-negative and takes no NSAIDs. His fasting serum gastrin is markedly elevated. His father had hyperparathyroidism and a pituitary adenoma.

Which test best confirms the suspected diagnosis?

A. Urea breath test
B. Serum chromogranin A alone
C. 24-hour urinary 5-HIAA
D. Secretin stimulation test
E. D-xylose absorption test

39. A 27-year-old woman with ulcerative colitis has a severe flare with 10 bloody stools per day. Her temperature is 39.0°C and heart rate 124/min, and her abdomen is distended and tender. Abdominal x-ray shows a transverse colon dilated to 8 cm.

Which medication should be avoided?

A. Loperamide
B. IV fluids
C. Broad-spectrum IV ABX
D. Subcutaneous heparin for DVT prophylaxis
E. IV methylprednisolone

40. A 71-year-old man who smokes has 3 months of progressive dysphagia to solids and liquids and has lost 11 kg. Barium swallow shows a "bird's beak" narrowing at the gastroesophageal junction.

Which is the most appropriate next step?

A. Upper endoscopy
B. Trial of a calcium channel blocker
C. Botulinum toxin injection
D. Pneumatic dilation
E. Esophageal manometry

41. A 23-year-old woman with acute liver failure after a paracetamol overdose becomes sweaty and less responsive in the ICU. Her INR is 4.2.

Which should be checked immediately?

A. Thyroid-stimulating hormone
B. Serum lipase
C. Blood glucose
D. Ceruloplasmin
E. Serum ferritin

42. A 63-year-old man has painless jaundice. Imaging shows strictures of the bile ducts and a diffusely enlarged, "sausage-shaped" pancreas. Serum IgG4 is markedly elevated, and no mass is seen.

Which is the most appropriate treatment?

A. Pancreaticoduodenectomy
B. Liver transplant
C. Corticosteroids
D. Ursodeoxycholic acid alone
E. Broad-spectrum ABX

43. A 40-year-old man has had severe watery diarrhea for 4 days. He is lethargic and has dry mucous membranes.

Which arterial blood gas pattern is most likely?

A. Low pH, low HCO_3^-, elevated anion gap, low $PaCO_2$
B. Low pH, low HCO_3^-, normal anion gap, low $PaCO_2$
C. Low pH, high HCO_3^-, high $PaCO_2$
D. High pH, low HCO_3^-, low $PaCO_2$
E. High pH, high HCO_3^-, high $PaCO_2$

44. A 49-year-old man with a long history of heavy alcohol use has greasy, foul-smelling stools and weight loss. Abdominal imaging shows pancreatic calcifications.

Which set of test results is most likely?

A. Low urinary D-xylose; normal fecal elastase
B. Positive anti-tTG; normal fecal elastase
C. Low urinary D-xylose; low fecal elastase
D. Normal urinary D-xylose; normal fecal elastase
E. Normal urinary D-xylose; low fecal elastase

45. A 52-year-old man is admitted with acute pancreatitis. Heart rate is 112/min, BUN is rising, and urine output is low.

Which is the most important initial treatment?

A. Total parenteral nutrition
B. Prophylactic IV carbapenem to prevent infected necrosis
C. Urgent contrast CT
D. Emergency necrosectomy
E. Goal-directed IV fluids such as Hartmann solution

46. A 59-year-old woman with long-standing autoimmune atrophic gastritis has persistently high serum gastrin.

Chronic hypergastrinemia puts her at increased risk of which tumor?

A. Cholangiocarcinoma
B. Pancreatic adenocarcinoma
C. Gastric carcinoid (neuroendocrine) tumor
D. Hepatocellular carcinoma
E. Esophageal squamous cell carcinoma

47. A 61-year-old woman with vitiligo and Hashimoto thyroiditis has fatigue and numbness in her feet. Hemoglobin is 9.1 g/dL with an MCV of 112 fL. Antibodies to parietal cells and intrinsic factor are positive.

Which region of the stomach is most likely to show atrophy?

A. Pylorus
B. Antrum
C. Cardia only
D. Body and fundus
E. Duodenal bulb

48. A medical student asks why vaccines exist for hepatitis A and B, but not for hepatitis C.

Which feature of HCV best explains the lack of a vaccine?

A. It is spread only by the fecal-oral route
B. High antigenic variation of its envelope glycoproteins
C. It can only replicate with another virus present
D. It is a DNA virus that integrates into the genome
E. It is a naked virus with no envelope

49. A 19-year-old woman is brought in 6 hours after taking 30 tablets of 500 mg paracetamol. She has mild nausea but is otherwise well. Her paracetamol level is above the treatment line on the nomogram.

The antidote she should receive works mainly by which mechanism?

A. Enhancing glucuronidation of paracetamol
B. Replenishing glutathione to neutralize NAPQI
C. Inhibiting CYP2E1
D. Binding paracetamol in the gut
E. Increasing renal excretion of paracetamol

50. A relief worker returning from an area hit by flooding develops profuse, painless "rice-water" diarrhea and becomes severely dehydrated within hours. Stool culture grows comma-shaped gram-negative rods.

Which mechanism is responsible for the diarrhea?

A. Mucosal invasion with neutrophilic infiltration
B. Toxin-mediated activation of guanylate cyclase only
C. Osmotic retention of water by unabsorbed sugars
D. Toxin-mediated ADP-ribosylation of Gs, increasing cAMP
E. Inactivation of the 60S ribosomal subunit

51. A 74-year-old woman with long-standing gallstones has air in the biliary tree on CT. She has no history of biliary surgery or ERCP.

Which type of biliary fistula is most common?

A. Cholecystoduodenal
B. Cholecystocolic
C. Cholecystogastric
D. Cholecystocutaneous
E. Choledochoduodenal

52. A 56-year-old woman with primary biliary cholangitis has fingers that turn white in the cold, tight skin over her fingers, small dilated blood vessels on her face, and difficulty swallowing.

Which antibody is most likely positive for her second condition?

A. Anti-Jo-1 antibody
B. Anti-CCP antibody
C. Anti-Scl-70 (topoisomerase I) antibody
D. Anti-dsDNA antibody
E. Anti-centromere antibody

53. A 35-year-old woman with constipation-predominant IBS has hard, infrequent stools and bloating despite increasing fiber and fluids. Her workup is normal.

Which is the most appropriate next treatment?

A. Codeine
B. Loperamide
C. Osmotic laxative such as polyethylene glycol
D. Diphenoxylate-atropine
E. Subtotal colectomy with ileorectal anastomosis

54. A 46-year-old woman with Raynaud phenomenon, sclerodactyly, and telangiectasias reports worsening heartburn and difficulty swallowing solids.

Which pair of manometry findings is most likely?

A. Normal LES pressure; normal peristalsis
B. Increased LES pressure; normal peristalsis
C. Normal LES pressure; simultaneous high-amplitude contractions
D. Increased LES pressure; absent distal peristalsis
E. Decreased LES pressure; decreased distal peristalsis

55. A 26-year-old woman with Crohn disease develops severe epigastric pain radiating to the back 4 weeks after starting a new medication. Lipase is 8 times normal. She does not drink alcohol, ultrasound shows no gallstones, and triglycerides and calcium are normal.

Which medication is the most likely cause?

A. Budesonide
B. Azathioprine
C. Loperamide
D. Vitamin D
E. Infliximab

56. A 26-year-old man with chronic RLQ pain, diarrhea, and weight loss has a small bowel follow-through showing a long, narrowed segment of the terminal ileum.

What is this radiologic finding called, and which disease does it suggest?

A. String sign; Crohn disease
B. Lead pipe colon; ulcerative colitis
C. Coffee bean sign; sigmoid volvulus
D. Apple core lesion; colorectal cancer
E. Bird's beak; achalasia

57. A 17-year-old girl has unexplained raised aminotransferases and a new hand tremor. Her uncle died of liver failure in his 20s.

Which combination of initial tests is most appropriate?

A. Alpha-1 antitrypsin level
B. Serum IgG4
C. Transferrin saturation and serum ferritin level
D. Serum ceruloplasmin and 24-hour urine copper
E. Anti-mitochondrial antibody

58. A 6-year-old boy with cystic fibrosis has bulky, greasy, foul-smelling stools and poor weight gain. Fecal elastase is very low.

Which is the most appropriate treatment for his steatorrhea?

A. Ursodeoxycholic acid
B. Lactase supplements
C. Gluten-free diet
D. Pancreatic enzyme replacement with meals
E. Oral vancomycin for Clostridioides difficile

59. A 68-year-old woman taking ibuprofen daily presents with melena and light-headedness. Blood pressure is 96/60 mm Hg and heart rate 112/min. Hemoglobin is 8.1 g/dL.

After resuscitation with IV fluids and blood, which is the most appropriate management?

A. Colonoscopy
B. Oral PPI and outpatient endoscopy in 4 weeks
C. IV PPI and upper endoscopy within 24 hours
D. Emergency laparotomy
E. Barium meal

60. A 34-year-old man with ulcerative colitis has an elevated ALP of 480 U/L on routine bloods, with normal bilirubin. Ultrasound shows no gallstones. He is p-ANCA positive and AMA negative.

Which test is most appropriate to confirm the diagnosis?

A. MRCP
B. Serum IgG4 alone
C. Liver biopsy
D. HIDA scan
E. Repeat ultrasound in 6 months

61. A 56-year-old man has 2 days of right upper quadrant pain, jaundice, dark urine, and itching. He is afebrile. Bilirubin is 5.8 mg/dL and ALP is 410 U/L, with mildly raised AST/ALT and normal lipase. Ultrasound shows gallstones and a common bile duct dilated to 10 mm.

Which is the most appropriate next step?

A. Laparoscopic cholecystectomy alone
B. CT-guided liver biopsy
C. HIDA scan
D. ERCP with stone extraction
E. Repeat LFTs in 2 weeks

62. A 68-year-old man who smokes has 6 weeks of progressive jaundice, itching, and a 7-kg weight loss, with no abdominal pain. On examination, he has a smooth, nontender mass below the right costal margin that moves with respiration. ALP and conjugated bilirubin are markedly raised.

Which is the most likely cause?

A. Choledocholithiasis
B. Mirizzi syndrome from a stone in the cystic duct
C. Pancreatic head or other periampullary malignancy
D. Hepatitis A
E. Acute cholecystitis with a stone in the gallbladder neck

63. A baby is born to a mother who is HBsAg positive. The baby is healthy, at term, and weighs 3.4 kg.

Which is the most appropriate management for the newborn?

A. Hepatitis B vaccine plus HBIG within 12 hours of birth
B. HBIG only
C. Tenofovir for the newborn for 6 months
D. Hepatitis B vaccine at 2 months of age
E. Test the newborn for HBsAg at birth and treat only if positive

64. A 71-year-old woman develops watery diarrhea 10 days after clindamycin for a dental abscess. She has 7 stools per day, a WBC of 13000/μL and normal creatinine. A stool test for C. difficile toxin is positive.

Which is the most appropriate treatment?

A. Loperamide
B. Observation only
C. Oral clindamycin
D. Oral vancomycin
E. IV vancomycin

65. A 38-year-old man who has never smoked has progressive dyspnea. Chest CT shows panacinar emphysema in the lower lobes. His liver enzymes are mildly elevated, and a liver biopsy shows PAS-positive, diastase-resistant globules in periportal hepatocytes.

Which mechanism best explains his liver disease?

A. Misfolded protein accumulating in the hepatocyte endoplasmic reticulum
B. Autoantibodies against bile duct epithelium
C. Copper deposition in hepatocytes
D. Unopposed neutrophil elastase digesting the connective tissue of the liver
E. Iron-mediated free radical damage

66. A 72-year-old woman with severe osteoarthritis needs to continue naproxen. She had an NSAID-related gastric erosion 2 years ago.

Which is the most appropriate way to reduce her GI risk while she continues the NSAID?

A. Add oral prednisone
B. Take the naproxen on an empty stomach
C. Add low-dose aspirin
D. Add an antacid only when symptomatic
E. Add a daily proton pump inhibitor

67. A 34-year-old woman taking combined oral contraceptives, who has had three pregnancies, is found to have cholesterol gallstones.

How does estrogen promote cholesterol gallstone formation?

A. It increases gallbladder contraction
B. It causes hemolysis
C. It increases cholesterol secretion into bile
D. It increases bile salt synthesis
E. It deconjugates bilirubin

68. A 52-year-old man has 2 days of left lower quadrant pain and a temperature of 37.9°C. He is tolerating fluids, has no peritoneal signs, and has no significant comorbidities. CT shows uncomplicated sigmoid diverticulitis without abscess or free air.

Which is the most appropriate management?

A. Outpatient clear liquid diet and close follow-up
B. Colonoscopy this week to exclude colorectal cancer
C. CT-guided drainage
D. Emergency sigmoid colectomy with colostomy
E. Admission for IV ABX, bowel rest, and TPN

69. A 38-year-old woman has a 3-cm liver lesion found by chance on ultrasound. Contrast MRI shows peripheral nodular enhancement that gradually fills in toward the center. She has no symptoms or liver disease.

Which is the most appropriate management?

A. Percutaneous biopsy
B. Surgical resection
C. Transarterial chemoembolization
D. Reassurance; no biopsy or treatment needed
E. Stop any oral contraceptives and repeat imaging in 1 month

70. A 28-year-old man had jaundice and fatigue 3 months ago after unprotected sex with a new partner. He now feels well. Serology shows: HBsAg negative, anti-HBs negative, HBeAg negative, anti-HBe positive, IgM anti-HBc positive.

Which phase of hepatitis B infection is he most likely in?

A. Successful vaccination
B. Window period
C. Chronic non-replicative infection
D. Complete recovery with immunity
E. Chronic replicative infection

71. A 24-year-old man is brought in 1 hour after intentionally drinking liquid drain cleaner. He is drooling and has oropharyngeal burns but is speaking clearly. Vital signs are stable, the abdomen is soft, and an upright chest x-ray shows no free air.

Which of the following should be avoided in this patient?

A. Assessing and securing the airway
B. Inducing emesis
C. Upper endoscopy within 24 hours to grade the injury
D. IV fluids
E. Keeping the patient NPO

72. A 60-year-old man with alcohol-related cirrhosis has moderate ascites. He has stopped drinking. Creatinine and sodium are normal.

Which is the most appropriate initial treatment?

A. Fluid restriction to 500 mL per day, with sodium intake left unrestricted
B. Furosemide alone
C. Immediate TIPS
D. Weekly large-volume paracentesis
E. Sodium restriction and spironolactone, with furosemide added as needed

73. A 29-year-old woman taking oral contraceptives is diagnosed with Budd-Chiari syndrome on Doppler ultrasound. She has mild ascites, a normal mental status, INR 1.3 and no evidence of cirrhosis.

Which is the most appropriate initial treatment?

A. Immediate liver transplantation
B. Observation with repeat ultrasound in 6 months
C. Transjugular intrahepatic portosystemic shunt (TIPS) before anticoagulation
D. Large-volume paracentesis alone
E. Anticoagulation and stopping the oral contraceptive

74. A 30-year-old pregnant woman with mild heartburn is started on famotidine.

Famotidine reduces acid secretion by which mechanism?

A. Blocking H2 receptors, lowering cAMP in parietal cells
B. Activating prostaglandin E1 receptors
C. Irreversibly blocking the proton pump in parietal cells
D. Blocking muscarinic M3 receptors
E. Binding bile salts

75. A 67-year-old woman with osteoporosis was started on weekly alendronate 3 weeks ago. She takes the tablet with a sip of water just before lying down for bed. She now reports sharp retrosternal pain when swallowing.

Which instruction would most likely have prevented her symptoms?

A. Take the tablet with a large meal so that the food carries it down into the stomach
B. Crush the tablet and mix it with food
C. Take the tablet with a full glass of water and stay upright for at least 30 minutes
D. Take the tablet with a glass of milk
E. Take the tablet with a calcium supplement

76. A 79-year-old woman has small bowel obstruction. CT shows air in the biliary tree, dilated small bowel, and a 3.5-cm gallstone in the distal ileum. She is resuscitated.

Which is the most appropriate surgical procedure?

A. Emergency cholecystectomy and fistula repair as the first step
B. Enterolithotomy to remove the stone from the bowel
C. Colonoscopic stone removal
D. Nasogastric decompression alone
E. Right hemicolectomy

77. A 28-year-old woman at 20 weeks' gestation has 1 day of abdominal pain, now worst in the right side of the abdomen, with nausea and a temperature of 38.0°C. WBC is 15000/μL.

Which is the most appropriate initial imaging?

A. No imaging; observe for 48 hours and repeat the white cell count
B. Graded-compression ultrasound, followed by MRI if inconclusive
C. Barium enema
D. CT AP with contrast
E. Abdominal x-ray

78. A 41-year-old man with known chronic hepatitis B, who injects drugs, has a sudden flare of hepatitis with jaundice. HBsAg is positive, IgM anti-HBc is negative, and IgG anti-HBc and anti-HDV are positive.

Which statement about his new infection is most accurate?

A. It is an HDV coinfection and will likely resolve
B. It can be prevented by the hepatitis C vaccine
C. It is an HDV superinfection and is likely to become chronic
D. It is reactivation of HBV alone
E. It is acute hepatitis A

79. A 75-year-old man with a common bile duct stone has fever, RUQ pain, and jaundice. Blood cultures are drawn before ABX are started.

Which organism is most likely to grow?

A. Staphylococcus aureus
B. Escherichia coli
C. Pseudomonas aeruginosa
D. Streptococcus pneumoniae
E. Candida albicans

80. A patient with a duodenal ulcer is prescribed a drug that forms a viscous, protective layer over the ulcer base. It must be taken on an empty stomach and works poorly when taken with a PPI.

Which drug is this?

A. Aluminum hydroxide
B. Misoprostol
C. Famotidine
D. Bismuth subsalicylate
E. Sucralfate

Test A answer sheet

Name _____ Date _____ Score _____ / 80

1 A B C D E	**21** A B C D E	**41** A B C D E	**61** A B C D E
2 A B C D E	**22** A B C D E	**42** A B C D E	**62** A B C D E
3 A B C D E	**23** A B C D E	**43** A B C D E	**63** A B C D E
4 A B C D E	**24** A B C D E	**44** A B C D E	**64** A B C D E
5 A B C D E	**25** A B C D E	**45** A B C D E	**65** A B C D E
6 A B C D E	**26** A B C D E	**46** A B C D E	**66** A B C D E
7 A B C D E	**27** A B C D E	**47** A B C D E	**67** A B C D E
8 A B C D E	**28** A B C D E	**48** A B C D E	**68** A B C D E
9 A B C D E	**29** A B C D E	**49** A B C D E	**69** A B C D E
10 A B C D E	**30** A B C D E	**50** A B C D E	**70** A B C D E
11 A B C D E	**31** A B C D E	**51** A B C D E	**71** A B C D E
12 A B C D E	**32** A B C D E	**52** A B C D E	**72** A B C D E
13 A B C D E	**33** A B C D E	**53** A B C D E	**73** A B C D E
14 A B C D E	**34** A B C D E	**54** A B C D E	**74** A B C D E
15 A B C D E	**35** A B C D E	**55** A B C D E	**75** A B C D E
16 A B C D E	**36** A B C D E	**56** A B C D E	**76** A B C D E
17 A B C D E	**37** A B C D E	**57** A B C D E	**77** A B C D E
18 A B C D E	**38** A B C D E	**58** A B C D E	**78** A B C D E
19 A B C D E	**39** A B C D E	**59** A B C D E	**79** A B C D E
20 A B C D E	**40** A B C D E	**60** A B C D E	**80** A B C D E

Test B

79 questions · 120 minutes

1. A 52-year-old woman has a positive anti-HCV antibody on routine one-time screening. She feels well, and her liver enzymes are normal.

Which is the most appropriate next step?

A. Start direct-acting antiviral therapy
B. Repeat anti-HCV antibody in 6 months
C. Reassure her that she is immune
D. HCV RNA by PCR
E. Liver biopsy

2. A 49-year-old woman with limited cutaneous systemic sclerosis (CREST) has frequent heartburn and regurgitation. Manometry shows low LES pressure and weak distal peristalsis.

Which is the most appropriate treatment for her esophageal symptoms?

A. Heller myotomy
B. Proton pump inhibitor
C. Calcium channel blocker
D. Botulinum toxin injection into the LES
E. Pneumatic dilation

3. A 45-year-old man with chronic hepatitis B and a high viral load is started on tenofovir. The drug targets a step in the viral life cycle that HBV shares with retroviruses.

Which step does tenofovir inhibit?

A. Cleavage of the viral polyprotein
B. Reverse transcription of pregenomic RNA into DNA
C. Entry through the NTCP receptor
D. Repair of pdsDNA into cccDNA
E. Transcription by host RNA polymerase II

4. A 78-year-old man has sudden, painless passage of a large volume of bright red blood per rectum. He is otherwise well, with a soft, nontender abdomen.

Which is the most likely cause?

A. Anal fissure
B. Diverticulosis
C. Ischemic colitis
D. Ulcerative colitis
E. Internal hemorrhoids

5. A 34-year-old man reports 6 months of abdominal discomfort and a change in bowel habits. He thinks he may have irritable bowel syndrome.

Which additional finding most strongly warrants further investigation before diagnosing IBS?

A. Symptoms that worsen during stressful periods
B. Iron-deficiency anemia
C. Mucus in the stool
D. Bloating that worsens through the day
E. Pain relieved by defecation

6. After an emergency delivery for acute fatty liver of pregnancy, the mother asks why the condition happened.

A deficiency in the fetus of which enzyme is most strongly associated?

A. Glucose-6-phosphatase
B. Hexosaminidase A
C. Medium-chain acyl-CoA dehydrogenase (MCAD)
D. Long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD)
E. Ornithine transcarbamylase

7. Two patients have chronic H. pylori infection. One strain carries a gene encoding a protein that is injected into gastric epithelial cells by a type IV secretion system.

Strains with this virulence factor carry a higher risk of which outcome?

A. Eosinophilic esophagitis
B. Gastric adenocarcinoma
C. Pancreatic insufficiency
D. Celiac disease
E. Achalasia

8. A 23-year-old woman who smokes has crampy right lower quadrant pain, non-bloody diarrhea, and weight loss. She also has a perianal fistula. Colonoscopy shows patchy ulcers in the terminal ileum and ascending colon separated by normal mucosa, and the rectum is spared.

Which biopsy finding is most likely?

A. PAS-positive foamy macrophages
B. Caseating granulomas
C. Crypt abscesses limited to the mucosa
D. Noncaseating granulomas with transmural inflammation
E. Villous atrophy with intraepithelial lymphocytes

9. A 23-year-old man had bloody diarrhea after eating undercooked chicken at a barbecue. Three weeks later he develops ascending, symmetric leg weakness and absent reflexes.

Which organism most likely caused his diarrhea?

A. Staphylococcus aureus
B. Rotavirus
C. Campylobacter jejuni
D. Giardia lamblia
E. Vibrio cholerae

10. A 44-year-old woman with severe acute pancreatitis develops perioral tingling and carpopedal spasm on hospital day 2.

Which mechanism best explains her symptoms?

A. Vitamin D toxicity
B. Calcium binding to free fatty acids (saponification)
C. Respiratory acidosis
D. Excess parathyroid hormone secretion
E. Increased renal calcium excretion from IV fluid loading

11. A 66-year-old man with painless jaundice has a CT showing a 2.5-cm mass in the head of the pancreas, with dilation of both the common bile duct and the pancreatic duct. There is no vascular invasion or metastasis.

Which is the most appropriate definitive treatment?

A. Observation with repeat CT in 3 months
B. Cholecystectomy
C. Pancreaticoduodenectomy (Whipple procedure)
D. Distal pancreatectomy
E. ERCP stent alone as definitive treatment

12. A 71-year-old woman has fever of 39.2°C, right upper quadrant pain, and jaundice. Over several hours she becomes confused, and her blood pressure falls to 82/48 mm Hg despite IV fluids. Ultrasound shows a dilated common bile duct with a stone.

Which is the most important definitive step after starting IV ABX and resuscitation?

A. HIDA scan
B. Oral ABX and outpatient follow-up
C. Urgent ERCP for biliary drainage
D. Ursodeoxycholic acid
E. Elective cholecystectomy in 6 weeks

13. A 31-year-old woman at 36 weeks' gestation has nausea, abdominal pain, and jaundice. Blood pressure is 128/80 mm Hg. Two possible diagnoses are being considered: acute fatty liver of pregnancy and HELLP syndrome.

Which finding most favors acute fatty liver of pregnancy?

A. Platelet count of 60000/μL with schistocytes
B. Raised LDH
C. Proteinuria
D. Hypoglycemia with prolonged INR
E. Blood pressure over 160/110 mm Hg

14. A 47-year-old man with chronic hepatitis B develops fever, weight loss, painful skin nodules, foot drop, and new hypertension. Angiography shows microaneurysms of the renal and mesenteric arteries.

Which is the most likely diagnosis?

A. Mixed cryoglobulinemia
B. Takayasu arteritis
C. Granulomatosis with polyangiitis
D. Polyarteritis nodosa
E. IgA vasculitis

15. A 29-year-old woman at 34 weeks' gestation has fever, malaise, and RUQ pain. AST is 4800 U/L and ALT 4200 U/L, but bilirubin is only mildly raised. Viral hepatitis A–E serologies are negative.

Which is the most appropriate next step while confirmatory testing is sent?

A. Start tenofovir
B. Start empiric IV acyclovir
C. Start ursodeoxycholic acid
D. Wait for liver biopsy results before treating
E. Start oral prednisone

16. A 68-year-old man with several prior episodes of sigmoid diverticulitis reports passing air bubbles in his urine and has had 3 urinary tract infections in 4 months.

Which is the most likely explanation for his symptoms?

A. Urethral stricture
B. Emphysematous cystitis
C. Colovesical fistula
D. Bladder exstrophy
E. Benign prostatic hyperplasia

17. A 62-year-old man who worked for decades in a plastics manufacturing plant presents with weight loss and hepatomegaly. Biopsy shows a malignant tumor of endothelial origin.

Which exposure most likely caused this tumor?

A. Aflatoxin
B. Vinyl chloride
C. Aniline dyes
D. Benzene
E. Asbestos

18. A 42-year-old woman has 1 year of progressive dysphagia to both solids and liquids, regurgitation of undigested food, and a 6-kg weight loss. Barium swallow shows a dilated esophagus that tapers to a narrow "bird's beak" at the gastroesophageal junction.

Which of the following best explains this condition?

A. Eosinophilic infiltration of the esophageal mucosa with subepithelial fibrosis and rings
B. Loss of ganglion cells in the submucosal (Meissner) plexus of the distal esophagus
C. Loss of inhibitory nitric oxide- and VIP-producing neurons in the myenteric plexus
D. Autoantibodies against the acetylcholine receptor
E. Fibrosis and atrophy of esophageal smooth muscle

19. A 45-year-old man from rural Southeast Asia, who often eats raw freshwater fish, presents with painless jaundice and weight loss. Imaging shows a mass at the confluence of the right and left hepatic ducts.

Which organism most likely increased his risk for this tumor?

A. Helicobacter pylori
B. Entamoeba histolytica
C. Hepatitis A virus
D. Schistosoma haematobium
E. Clonorchis sinensis

20. A 24-year-old nursing student has pre-placement blood tests. Results: HBsAg negative, anti-HBs positive, anti-HBc negative, HBeAg negative, anti-HBe negative.

Which best explains these results?

A. Immunity from vaccination
B. Chronic infection
C. Immunity from past natural infection
D. Acute infection
E. Window period

21. A 46-year-old man has 12 hours of severe epigastric pain radiating to his back, with vomiting. He drinks heavily. Lipase is 5 times the upper limit of normal. He is hemodynamically stable.

Which is the most appropriate next step to confirm the diagnosis?

A. Contrast CT abdomen immediately
B. Repeat lipase in 24 hours
C. No further test is needed
D. Endoscopic ultrasound
E. MRCP

22. A 41-year-old man has had burning retrosternal discomfort after large meals and when lying down for 3 months. He has no dysphagia, weight loss, vomiting, anemia, or GI bleeding. An ECG done for the discomfort was normal.

Which of the following is the most appropriate next step?

A. 24-hour ambulatory pH monitoring
B. 8-week trial of a proton pump inhibitor
C. Laparoscopic Nissen fundoplication
D. Barium swallow
E. Upper endoscopy with biopsy

23. A 55-year-old woman has intermittent episodes of severe chest pain and dysphagia, often triggered by very hot or cold drinks. A cardiac workup is negative. Manometry shows frequent simultaneous, uncoordinated contractions in the distal esophagus with normal LES relaxation.

Which barium swallow finding is most likely?

A. "Corkscrew" esophagus
B. Concentric rings ("feline" esophagus)
C. "Apple core" lesion
D. "Bird's beak" narrowing at the GE junction
E. Outpouching above the upper esophageal sphincter

24. A 22-year-old man has 18 hours of abdominal pain that began around the umbilicus and moved to the right lower quadrant, with anorexia and a low-grade fever. Lying on his left side, he has pain when his right hip is passively extended.

This finding suggests the inflamed appendix lies in which position?

A. Anterior, just under McBurney point
B. Left lower quadrant
C. Pelvic, against the obturator internus
D. Retrocecal, against the iliopsoas muscle
E. Subhepatic

25. A 42-year-old woman with obesity has had three episodes of epigastric pain radiating to her back, each starting about an hour after a large fried meal and settling within 3 hours. She is pain-free today, afebrile, with a soft, nontender abdomen. LFTs and white cell count are normal. Ultrasound shows several mobile, shadowing gallstones and a normal gallbladder wall.

Which is the most appropriate management?

A. No treatment; reassure
B. IV ABX and admission for emergency surgery
C. HIDA scan
D. ERCP with sphincterotomy
E. Elective laparoscopic cholecystectomy

26. A 24-year-old woman is stabbed in the right upper abdomen. Her blood pressure is 78/40 mm Hg and heart rate 136/min despite 2 units of blood. A FAST exam shows free fluid in Morison pouch.

Which is the most appropriate next step?

A. Repeat FAST in 30 minutes
B. CT abdomen with IV contrast
C. Angioembolization
D. Serial abdominal examinations
E. Emergency laparotomy

27. A 35-year-old previously healthy man develops jaundice 10 days after starting a new herbal supplement. He is now confused, with an INR of 2.4.

Which combination of features defines acute liver failure?

A. Encephalopathy and INR of 1.5 or more, with known cirrhosis
B. Jaundice and AST above 1000 U/L within 26 weeks, with no prior liver disease
C. Encephalopathy and INR of 1.5 or more within 26 weeks, with no prior liver disease
D. Ascites and splenomegaly
E. Bilirubin above 10 mg/dL alone

28. A 9-year-old girl has 1 day of periumbilical pain that has moved to the right lower quadrant, with vomiting and a temperature of 38.1°C. She is tender at McBurney point. WBC is 14500/µL.

Which is the most appropriate initial imaging test?

A. Barium enema
B. CT AP with contrast
C. Right lower quadrant ultrasound
D. Colonoscopy
E. Erect and supine abdominal x-ray

29. A 19-year-old man with asthma and atopic dermatitis presents to the emergency department with a piece of steak stuck in his throat. He reports years of intermittent dysphagia to solids. Endoscopy shows concentric esophageal rings and linear furrows. Biopsy shows more than 15 eosinophils per high-power field.

Which of the following is the most likely diagnosis?

A. Eosinophilic esophagitis
B. Candida esophagitis
C. Schatzki ring
D. Achalasia
E. Gastroesophageal reflux disease

30. A 46-year-old man with long-standing primary sclerosing cholangitis and ulcerative colitis develops worsening jaundice, weight loss, and a new dominant stricture at the hepatic duct bifurcation. CA 19-9 is markedly elevated.

Which is the most likely diagnosis?

A. Pancreatic pseudocyst
B. Cholangiocarcinoma
C. Choledocholithiasis
D. Hepatic adenoma
E. Primary biliary cholangitis

31. A 58-year-old woman with chronic dyspepsia undergoes endoscopy. Gastric biopsy shows a low-grade, localized marginal-zone B-cell lymphoma of mucosa-associated lymphoid tissue. H. pylori is present on histology.

Which is the most appropriate initial treatment?

A. H. pylori eradication therapy
B. R-CHOP chemotherapy
C. Radiation therapy
D. Observation without treatment
E. Total gastrectomy

32. A 48-year-old man is diagnosed with HFE hereditary hemochromatosis. His hemoglobin is 15.2 g/dL, ferritin is 1600 ng/mL, and he has no cardiac disease.

Which is the most appropriate treatment?

A. Low-iron diet alone
B. Liver transplant
C. Therapeutic phlebotomy
D. D-penicillamine
E. Oral deferasirox

33. A 41-year-old man has abdominal distention and RUQ pain. Doppler ultrasound shows thrombosis of the hepatic veins. His hemoglobin is 19.2 g/dL, and he complains of itching after hot showers.

Which test is most likely to reveal the underlying cause?

A. JAK2 V617F mutation testing
B. Anti-mitochondrial antibody
C. Hepatitis B surface antigen
D. Serum ceruloplasmin
E. Alpha-1 antitrypsin level

34. A 42-year-old woman had 100 cm of terminal ileum resected for Crohn disease. Several years later she has fatigue, glossitis, and a macrocytic anemia.

Deficiency of which nutrient is most likely?

A. Vitamin B12
B. Vitamin C
C. Folate
D. Iron
E. Zinc

35. A gastric biopsy from a patient with a duodenal ulcer shows curved, flagellated, gram-negative rods in the mucus layer overlying the antral epithelium.

Which enzyme allows this organism to survive in the acidic gastric environment?

A. Urease
B. Coagulase
C. Catalase
D. IgA protease
E. Hyaluronidase

36. A 44-year-old man with PiZZ alpha-1 antitrypsin deficiency has progressive dyspnea and panacinar emphysema in the lower lobes. He still smokes.

Which management most directly addresses his lung disease?

A. Therapeutic phlebotomy
B. Copper chelation
C. Ursodeoxycholic acid
D. Liver transplant to correct the underlying protein defect
E. Smoking cessation and IV A1AT augmentation

37. A 45-year-old woman with dyspepsia has a positive urea breath test. She has no drug allergies and has never taken macrolide ABX. Local clarithromycin resistance is low.

Which regimen is most appropriate?

A. H2-receptor antagonist + metronidazole
B. PPI + amoxicillin + clarithromycin
C. Sucralfate + misoprostol
D. Amoxicillin alone for 14 days
E. PPI alone for 8 weeks

38. A 59-year-old woman with septic shock from pyelonephritis becomes normotensive after IV fluids, noradrenaline, and ABX. On day 2, AST is 3100 U/L and ALT is 2700 U/L; INR is 1.4 and she is alert. Viral hepatitis serologies and the paracetamol level are negative.

Which is the most appropriate next step in managing her liver injury?

A. Urgent referral for liver transplantation
B. Percutaneous liver biopsy to exclude autoimmune hepatitis
C. Ursodeoxycholic acid
D. Maintain perfusion, treat the infection, and trend LFTs
E. IV methylprednisolone

39. A 17-year-old girl started doxycycline for acne 10 days ago. She takes it with a sip of water right before bed. She now has sharp retrosternal pain when swallowing. She has no history of heartburn and is otherwise well.

Which is the most likely diagnosis?

A. Achalasia
B. Eosinophilic esophagitis
C. Gastroesophageal reflux disease
D. Candida esophagitis
E. Pill-induced esophagitis

40. A 38-year-old woman has dysphagia to solids and liquids and occasional regurgitation. Barium swallow suggests achalasia, and upper endoscopy shows no mass at the gastroesophageal junction.

Which test is the gold standard for confirming the diagnosis?

A. Esophageal manometry
B. Endoscopic ultrasound
C. CT chest
D. 24-hour esophageal pH monitoring
E. Repeat barium swallow

41. A 62-year-old woman has gallstones on ultrasound and recent RUQ pain. ALP and GGT are mildly raised, bilirubin is 1.8 mg/dL, and the common bile duct measures 7 mm. No stone is seen in the duct. She has no fever.

Which is the most appropriate next step?

A. Immediate ERCP
B. Repeat LFTs in 6 months
C. HIDA scan
D. Cholecystectomy without further imaging
E. MRCP or endoscopic ultrasound

42. A 69-year-old man in the ICU with cardiogenic shock is receiving high-dose noradrenaline. He develops abdominal distention and rising lactate. CT angiography shows diffuse narrowing of the mesenteric arteries without an embolus or thrombus.

Which is the most likely diagnosis?

A. Mesenteric venous thrombosis
B. Chronic mesenteric ischemia
C. SMA embolus
D. Volvulus
E. Non-occlusive mesenteric ischemia

43. A 44-year-old woman with GERD is started on omeprazole taken 30 minutes before breakfast.

Omeprazole works by which mechanism?

A. Competitive blockade of H2 receptors on parietal cells
B. Increasing LES tone
C. Irreversible inhibition of the H+/K+ ATPase in parietal cells
D. Neutralizing gastric acid in the lumen
E. Forming a protective barrier over the mucosa

44. A 70-year-old man has crampy left-sided abdominal pain followed by bloody diarrhea 1 day after an episode of hypotension during dialysis. He is hemodynamically stable without peritonitis. CT shows thickening of the splenic flexure.

Which is the most appropriate initial management?

A. Oral vancomycin
B. Supportive care: bowel rest, IV fluids, and correcting hypotension
C. Emergency subtotal colectomy
D. High-dose IV corticosteroids
E. IV heparin and catheter-directed thrombolysis of the affected artery

45. A 61-year-old man has had heartburn for years. Over the past 2 months he has had difficulty swallowing solid food and has lost 6 kg. His hemoglobin is 10.8 g/dL with a low MCV.

Which is the most appropriate next step?

A. 24-hour pH monitoring
B. Upper endoscopy with biopsy
C. H. pylori stool antigen test
D. 8-week trial of a PPI
E. Esophageal manometry

46. A 77-year-old man is in the ICU on vasopressors for septic shock. Two days later he develops crampy left-sided abdominal pain followed by bloody diarrhea.

Which region of the colon is most vulnerable to this injury?

A. Terminal ileum
B. Mid-transverse colon
C. Cecum and ascending colon
D. Distal rectum
E. Splenic flexure

47. A 52-year-old woman has chronic watery diarrhea. Stool osmotic gap is low. Colonoscopy shows dark brown pigmentation of the colonic mucosa, and biopsy shows pigment-laden macrophages in the lamina propria.

Which is the most likely cause of her diarrhea?

A. Lactose intolerance
B. Surreptitious polyethylene glycol use
C. Celiac disease
D. Surreptitious stimulant laxative use
E. Vipoma

48. A 61-year-old man using an inhaled corticosteroid for COPD and recently treated with broad-spectrum ABX has painful swallowing. Endoscopy shows adherent white plaques along the esophagus.

Which finding would most likely be seen on a biopsy of the plaques?

A. Owl's eye inclusions in endothelial cells
B. More than 15 eosinophils per high-power field
C. Pseudohyphae and budding yeast
D. Goblet cells in columnar epithelium
E. Multinucleated giant cells

49. A 70-year-old man taking ibuprofen daily for knee osteoarthritis develops epigastric pain. Endoscopy shows multiple shallow erosions in the gastric antrum.

Reduced synthesis of which substance is primarily responsible for this injury?

A. Prostaglandin E2
B. Histamine
C. Gastrin
D. Somatostatin
E. Leukotriene B4

50. A 45-year-old man with Crohn disease, maintained on azathioprine, is started on allopurinol for gout. Three weeks later he has severe pancytopenia.

Which mechanism best explains this interaction?

A. Inhibition of CYP3A4
B. Displacement of azathioprine from albumin
C. Inhibition of xanthine oxidase, which normally breaks down 6-mercaptopurine
D. Induction of thiopurine methyltransferase, which normally breaks down 6-mercaptopurine
E. Increased renal clearance of azathioprine

51. A 54-year-old man has fatigue, joint pain in his hands, and new-onset diabetes. His skin is diffusely tanned and his liver is enlarged. Transferrin saturation is 82% and ferritin is 2400 ng/mL. He is homozygous for the HFE C282Y mutation.

Which mechanism best explains his iron overload?

A. Decreased ceruloplasmin activity
B. Increased hepcidin, leading to ferroportin degradation
C. Increased transferrin receptor expression on red blood cells
D. Decreased hepcidin, leading to increased ferroportin activity
E. Repeated blood transfusions

52. A patient with autoimmune atrophic gastritis undergoes laboratory testing.

Which set of findings is expected?

A. Increased gastric acid; increased serum gastrin
B. Normal gastric acid; normal serum gastrin
C. Increased gastric acid; decreased serum gastrin
D. Decreased gastric acid; increased serum gastrin
E. Decreased gastric acid; decreased serum gastrin

53. A 61-year-old woman has recovered from her first episode of CT-confirmed sigmoid diverticulitis. She has never had a colonoscopy.

Which is the most appropriate follow-up?

A. Colonoscopy during the acute episode
B. No follow-up is needed
C. Repeat CT every 6 months
D. Elective sigmoid colectomy
E. Colonoscopy 6–8 weeks after recovery

54. A 49-year-old woman is diagnosed with primary biliary cholangitis. She has a markedly elevated ALP, positive AMA, and no cirrhosis.

Which is the most appropriate first-line treatment to slow disease progression?

A. Prednisone
B. Methotrexate
C. Referral for liver transplant
D. Ursodeoxycholic acid
E. Cholestyramine

55. A 30-year-old man is diagnosed with primary sclerosing cholangitis after MRCP shows beaded intra- and extrahepatic bile ducts. He has no GI symptoms.

Which condition should he be evaluated for?

A. Celiac disease
B. Hashimoto thyroiditis
C. Hemochromatosis
D. Wilson disease
E. Ulcerative colitis

56. A 22-year-old woman presents 1 hour after swallowing 40 tablets of 500 mg paracetamol in a suicide attempt. She weighs 60 kg, is alert, and has no symptoms.

Which is the most appropriate next step?

A. Liver transplant referral
B. Induce vomiting
C. Measure LFTs now and treat only if they are abnormal
D. Discharge with outpatient follow-up because she has no symptoms
E. Activated charcoal, and measure a paracetamol level 4 hours after ingestion

57. A 30-year-old man develops bloating, foul-smelling greasy stools, and flatulence 10 days after a backpacking trip during which he drank untreated stream water. He has no fever or blood in the stool.

Which organism is the most likely cause?

A. Campylobacter jejuni
B. Shigella sonnei
C. Entamoeba histolytica
D. Norovirus
E. Giardia lamblia

58. A 41-year-old woman with a positive H. pylori stool antigen test had anaphylaxis to penicillin as a child. Local clarithromycin resistance is low.

Which triple therapy regimen is most appropriate?

A. PPI + clarithromycin + metronidazole
B. PPI + amoxicillin + rifabutin
C. PPI alone for 8 weeks
D. PPI + amoxicillin + levofloxacin
E. PPI + amoxicillin + clarithromycin

59. An epidemiology lecture compares the risk factors for Crohn disease and ulcerative colitis.

Which risk factor has opposite associations with the two diseases?

A. Northern European ancestry
B. Recent ABX use
C. Cigarette smoking
D. Western diet
E. Family history

60. A 29-year-old man with moderate-to-severe Crohn disease that has not responded to corticosteroids and azathioprine is to start infliximab.

Which test should be performed before starting therapy?

A. Interferon-gamma release assay (or tuberculin skin test)
B. Serum ceruloplasmin
C. Thiopurine methyltransferase (TPMT) enzyme activity level
D. Pulmonary function tests
E. Anti-Saccharomyces cerevisiae antibodies

61. A 14-month-old unvaccinated child who attends childcare has 3 days of vomiting, fever, and profuse watery diarrhea, and is moderately dehydrated.

Which organism is the most likely cause?

A. Norovirus
B. Rotavirus
C. Vibrio cholerae
D. Campylobacter jejuni
E. Giardia lamblia

62. A 45-year-old man with central obesity and prediabetes is found to have MASLD. FIB-4 shows a low risk of advanced fibrosis.

Which is the most appropriate initial management?

A. Ursodeoxycholic acid
B. Liver biopsy
C. Therapeutic phlebotomy
D. Referral for liver transplant evaluation
E. Lifestyle changes aiming for 7–10% weight loss

63. A 57-year-old man with untreated hereditary hemochromatosis develops fever, hypotension, and hemorrhagic bullae on his legs 1 day after eating raw oysters.

Which organism is most likely responsible?

A. Staphylococcus aureus
B. Vibrio cholerae
C. Bacillus cereus
D. Vibrio vulnificus
E. Listeria monocytogenes

64. A 64-year-old woman with cirrhosis has had two admissions for hepatic encephalopathy in 4 months despite taking lactulose regularly, with 2–3 soft stools a day.

Which is the most appropriate addition to her treatment?

A. A strict protein-free diet
B. Rifaximin
C. Metronidazole for life
D. Loperamide
E. Lorazepam at night

65. A 68-year-old man has tender hepatomegaly, ascites, bilateral leg edema, and distended neck veins that rise further with pressure over the liver. Bilirubin and AST/ALT are mildly raised.

Which is the most appropriate next test?

A. Anti-smooth muscle antibody
B. Liver biopsy
C. Doppler ultrasound of the hepatic veins
D. Echocardiography
E. MRCP

66. A 26-year-old woman has bloating, foul-smelling stools, iron-deficiency anemia, and an itchy vesicular rash on her elbows and knees. Duodenal biopsy shows villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes.

This condition is most strongly associated with which HLA types?

A. HLA-DQ2 and DQ8
B. HLA-DR5
C. HLA-B27
D. HLA-DR3 and DR4
E. HLA-DR2

67. A 26-year-old woman develops watery diarrhea 3 days after arriving in Mexico. She has 4 loose stools a day, no fever and no blood in the stool, and is drinking well.

Which is the most appropriate management?

A. Stool ova and parasites, then empirical metronidazole
B. Oral rehydration, with loperamide if needed
C. IV ciprofloxacin and admission
D. Oral vancomycin
E. Urgent colonoscopy

68. A 66-year-old man with rheumatoid arthritis needs long-term NSAID therapy and has a history of a gastric ulcer. He is started on a drug that is a prostaglandin E1 analog to protect the gastric mucosa.

This drug is contraindicated in which group of patients?

A. Patients with asthma
B. Patients with renal impairment
C. Patients taking warfarin
D. Pregnant patients
E. Patients with hypothyroidism

69. An 81-year-old woman with long-standing gallstones has chronic watery diarrhea and weight loss. Imaging shows air in the biliary tree and an abnormal connection between the gallbladder and the hepatic flexure of the colon.

Which best explains her diarrhea?

A. Bile salts entering the colon directly
B. Lactose intolerance
C. Pancreatic enzyme deficiency
D. Bacterial overgrowth in the stomach
E. Small bowel obstruction

70. A 33-year-old woman with diarrhea-predominant IBS and persistent abdominal pain is started on low-dose amitriptyline at night.

Which adverse effects is she most likely to experience?

A. Diarrhea and hypersalivation
B. Hyperkalemia and gynecomastia
C. Bradycardia and miosis
D. Weight loss and insomnia
E. Dry mouth and constipation

71. A 19-year-old man with chronic diarrhea and iron-deficiency anemia has a negative IgA anti-tTG test. His total serum IgA is undetectable, and he has a history of recurrent sinus infections.

Which is the most appropriate next step to evaluate for celiac disease?

A. Hydrogen breath test
B. Conclude celiac disease is excluded
C. Repeat IgA anti-tTG in 6 months
D. Fecal elastase
E. IgG-based celiac serology

72. A 52-year-old man with cirrhosis vomits a large amount of bright red blood. His blood pressure is 88/54 mm Hg. Two large-bore IV lines are placed and transfusion has started.

In addition to resuscitation, which drug combination should be started before endoscopy?

A. IV octreotide and IV ABX
B. Oral propranolol and lactulose
C. IV vitamin K alone
D. IV furosemide and spironolactone
E. IV pantoprazole alone

73. A 64-year-old woman with autoimmune gastritis has megaloblastic anemia, low serum B12, and reduced vibration sense in her feet.

Which is the most appropriate treatment?

A. Oral folic acid alone
B. Oral iron
C. Intramuscular vitamin B12
D. Proton pump inhibitor
E. H. pylori eradication therapy

74. A 58-year-old man with alcohol-related cirrhosis and ascites has fever, diffuse abdominal tenderness, and new confusion. Diagnostic paracentesis shows an ascitic neutrophil count of 520/µL.

Which is the most appropriate treatment?

A. IV third-generation cephalosporin plus IV albumin
B. Large-volume paracentesis with albumin, without ABX
C. Oral rifaximin alone
D. Emergency laparotomy
E. Increase the diuretic dose

75. A 22-year-old woman has bloating, flatulence, and loose stools within a few hours of drinking milk. A hydrogen breath test after a lactose load is positive.

Which stool finding is most likely during symptoms?

A. Decreased stool osmotic gap
B. Many fecal leukocytes
C. Increased stool osmotic gap
D. PAS-positive macrophages
E. Elevated fecal calprotectin

76. Two hours after eating leftover fried rice from a takeaway shop, a 25-year-old man develops nausea and repeated vomiting. His symptoms resolve within 24 hours.

Which organism is most likely responsible?

A. Bacillus cereus
B. Staphylococcus aureus
C. Salmonella enteritidis
D. Enterotoxigenic E. coli
E. Clostridium perfringens

77. A 48-year-old woman has 18 hours of constant right upper quadrant pain radiating to her right shoulder, with nausea. Temperature is 38.4°C. She stops breathing in when the right upper quadrant is palpated. WBC is 15800/µL; bilirubin, ALP, and lipase are normal. Ultrasound shows gallstones, a 5-mm gallbladder wall, and fluid around the gallbladder.

Which is the most appropriate definitive management?

A. Oral ursodeoxycholic acid
B. ERCP
C. Outpatient follow-up in 6 weeks
D. IV ABX and laparoscopic cholecystectomy during this admission
E. Percutaneous cholecystostomy as first-line in all patients

78. A 50-year-old woman has sudden severe epigastric pain radiating to her back with vomiting. Lipase is 9 times the upper limit of normal, ALT is 320 U/L, and ALP is raised. She drinks alcohol rarely. Ultrasound shows gallstones.

Where is the obstructing stone most likely located?

A. Gallbladder fundus
B. Cystic duct
C. Hepatopancreatic ampulla (of Vater)
D. Right hepatic duct at the porta hepatis
E. Common hepatic duct near the hilum

79. A biochemistry lecture explains why chronic alcohol use leads to fat accumulation in hepatocytes, regardless of whether alcohol is broken down by alcohol dehydrogenase, CYP2E1, or catalase.

Which change most directly promotes fatty liver?

A. Increased fatty acid beta-oxidation
B. Increased gluconeogenesis
C. Decreased glycerol-3-phosphate
D. Increased NADH:NAD+ ratio
E. Decreased acetaldehyde production

Test B answer sheet

Name _____ Date _____ Score _____ / 79

1 A B C D E	**21** A B C D E	**41** A B C D E	**61** A B C D E
2 A B C D E	**22** A B C D E	**42** A B C D E	**62** A B C D E
3 A B C D E	**23** A B C D E	**43** A B C D E	**63** A B C D E
4 A B C D E	**24** A B C D E	**44** A B C D E	**64** A B C D E
5 A B C D E	**25** A B C D E	**45** A B C D E	**65** A B C D E
6 A B C D E	**26** A B C D E	**46** A B C D E	**66** A B C D E
7 A B C D E	**27** A B C D E	**47** A B C D E	**67** A B C D E
8 A B C D E	**28** A B C D E	**48** A B C D E	**68** A B C D E
9 A B C D E	**29** A B C D E	**49** A B C D E	**69** A B C D E
10 A B C D E	**30** A B C D E	**50** A B C D E	**70** A B C D E
11 A B C D E	**31** A B C D E	**51** A B C D E	**71** A B C D E
12 A B C D E	**32** A B C D E	**52** A B C D E	**72** A B C D E
13 A B C D E	**33** A B C D E	**53** A B C D E	**73** A B C D E
14 A B C D E	**34** A B C D E	**54** A B C D E	**74** A B C D E
15 A B C D E	**35** A B C D E	**55** A B C D E	**75** A B C D E
16 A B C D E	**36** A B C D E	**56** A B C D E	**76** A B C D E
17 A B C D E	**37** A B C D E	**57** A B C D E	**77** A B C D E
18 A B C D E	**38** A B C D E	**58** A B C D E	**78** A B C D E
19 A B C D E	**39** A B C D E	**59** A B C D E	**79** A B C D E
20 A B C D E	**40** A B C D E	**60** A B C D E	

Test C

79 questions · 120 minutes

1. A 61-year-old man with HCV cirrhosis has a 3-cm liver lesion on surveillance ultrasound. Multiphase CT shows arterial-phase enhancement with washout in the portal venous phase. AFP is 420 ng/mL.

Which is the most appropriate next step to confirm the diagnosis?

A. PET scan
B. Percutaneous biopsy
C. No further test is needed
D. Diagnostic laparoscopy
E. Repeat ultrasound in 3 months

2. An 86-year-old woman with achalasia and severe heart failure is not a candidate for dilation or surgery. She receives an endoscopic injection into the lower esophageal sphincter.

The injected agent works by which mechanism?

A. Stimulating muscarinic receptors
B. Blocking acetylcholine release from nerve terminals
C. Blocking L-type calcium channels in smooth muscle
D. Inhibiting the proton pump
E. Releasing nitric oxide

3. A 34-year-old man has 6 days of right lower quadrant pain and fever. He is hemodynamically stable, without generalized peritonitis. CT shows a perforated appendix with a 5-cm well-defined abscess.

Which is the most appropriate management?

A. Colonoscopy to exclude malignancy before any treatment
B. Immediate open appendectomy
C. IV ABX and percutaneous drainage of the abscess
D. Oral ABX and discharge
E. Observation without ABX

4. A 45-year-old man's brother was recently diagnosed with HFE hereditary hemochromatosis. He feels well and asks to be tested.

Which is the best initial test?

A. Transferrin saturation and serum ferritin
B. Echocardiography
C. MRI of the liver
D. Liver biopsy with Prussian blue stain
E. Serum ceruloplasmin

5. A 69-year-old man with acute mesenteric ischemia becomes hypotensive while awaiting surgery.

Which acid-base disturbance is most likely?

A. Metabolic alkalosis from vomiting
B. Respiratory alkalosis alone
C. Non-anion-gap metabolic acidosis from bicarbonate loss
D. Respiratory acidosis from hypoventilation
E. Anion-gap metabolic acidosis from lactate

6. A 54-year-old man has months of diarrhea, weight loss, migratory arthralgias, and recent memory problems. Small-bowel biopsy shows the lamina propria packed with foamy macrophages containing PAS-positive material.

Which organism is the most likely cause?

A. Cryptosporidium parvum
B. Giardia lamblia
C. Klebsiella pneumoniae
D. Tropheryma whipplei
E. Mycobacterium avium complex

7. A 50-year-old man who emigrated from rural Bolivia has dysphagia to solids and liquids. Barium swallow shows a "bird's beak" appearance. Echocardiography shows a dilated left ventricle with an apical aneurysm.

Which organism is the most likely cause?

A. Trypanosoma brucei
B. Toxoplasma gondii
C. Trypanosoma cruzi
D. Leishmania donovani
E. Trichinella spiralis

8. A pathology lecture describes diverticulosis in Western patients.

Which part of the colon is most commonly affected, and why?

A. Ascending colon, because it has the thinnest muscular wall of the colon
B. Rectum, because it lacks a muscularis propria
C. Sigmoid colon, because it has the highest intraluminal pressure
D. Cecum, because it has the widest diameter
E. Transverse colon, because it lacks taeniae coli

9. A 66-year-old man in the ICU after major burns, on TPN and vasopressors, develops fever and leukocytosis. Ultrasound shows a distended gallbladder with a thickened wall and fluid around it, but no gallstones. He is too unstable for surgery.

Which is the most appropriate management?

A. Oral ursodeoxycholic acid
B. ERCP with sphincterotomy
C. IV ABX and percutaneous cholecystostomy
D. Observation only
E. Elective cholecystectomy in 6 weeks

10. A 22-year-old woman is diagnosed with Wilson disease with hepatic and mild neurologic involvement. She has no signs of acute liver failure.

Which is the most appropriate initial treatment?

A. Copper chelation with trientine or D-penicillamine
B. Immediate referral for urgent liver transplant evaluation
C. Iron chelation with deferoxamine or deferasirox
D. Therapeutic phlebotomy
E. Ursodeoxycholic acid

11. A 24-year-old man with allergic rhinitis is diagnosed with eosinophilic esophagitis after an episode of food impaction. He asks how to address the underlying cause of his condition.

Which management targets the trigger of this condition?

A. Pneumatic dilation of the LES
B. Long-term oral fluconazole
C. Laparoscopic Nissen fundoplication for reflux
D. Stopping all oral medications for 2 weeks
E. Elimination of food allergens from the diet

12. Two patients have AST and ALT above 3000 U/L. One has acute viral hepatitis; the other had a cardiac arrest 24 hours ago.

Which finding most favors ischemic hepatitis?

A. AST:ALT ratio above 2 with AST under 300
B. Positive IgM anti-HAV
C. Very high LDH, with aminotransferases that fall rapidly over days
D. Gradual rise in aminotransferases over 2 weeks, with a slow fall over months
E. Markedly elevated ALP with normal bilirubin

13. A 40-year-old man has 3 months of burning epigastric pain that begins 2–3 hours after meals and often wakes him at night. The pain improves when he eats, and he has gained 3 kg.

Which of the following is the most likely diagnosis?

A. Duodenal ulcer
B. Gastric ulcer
C. Chronic pancreatitis
D. Gastric adenocarcinoma
E. Biliary colic

14. A 66-year-old man is in the ICU on mechanical ventilation for 4 days after a severe head injury. His platelet count is 48000/μL.

Which intervention best reduces his risk of stress-related gastric bleeding?

A. Prophylactic PPI or H2 blocker
B. No intervention is needed
C. H. pylori testing
D. Routine daily upper endoscopy
E. Prophylactic misoprostol

15. A 50-year-old woman with chronic hepatitis C is started on a regimen that includes sofosbuvir.

Sofosbuvir acts by inhibiting which viral protein?

A. RNA-dependent DNA polymerase (reverse transcriptase)
B. Viral integrase
C. Neuraminidase
D. NS5B RNA-dependent RNA polymerase
E. NS3/4A protease

16. A 34-year-old man who recently returned from India has bloody diarrhea and, a few weeks later, fever with right upper quadrant pain. CT shows a single liver abscess. Stool microscopy shows trophozoites containing ingested red blood cells.

Which is the most appropriate treatment?

A. Metronidazole, then luminal paromomycin
B. Ciprofloxacin and percutaneous abscess drainage
C. Oral vancomycin
D. Albendazole, then surgical drainage
E. Oral rehydration only

17. A 38-year-old woman with untreated celiac disease and steatorrhea notices easy bruising and prolonged bleeding after a dental extraction.

Which laboratory pattern is most likely?

A. Prolonged PT; normal platelet count
B. Normal PT; low platelet count
C. Prolonged bleeding time; normal PT
D. Isolated prolonged PTT that corrects with mixing; low factor VIII
E. Normal PT and PTT; low fibrinogen

18. A 55-year-old woman with primary biliary cholangitis on ursodeoxycholic acid has severe itching that keeps her awake. She is started on a drug that binds bile salts in the intestine and prevents their reabsorption.

Which drug was started?

A. Orlistat
B. Polyethylene glycol
C. Ezetimibe
D. Cholestyramine
E. Octreotide

19. A 14-year-old boy with sickle cell disease has recurrent right upper quadrant pain after meals. An abdominal x-ray shows several radiopaque stones in the right upper quadrant.

Which best explains the formation of these stones?

A. Increased unconjugated bilirubin from chronic hemolysis
B. Calcium oxalate precipitation
C. Bacterial deconjugation of bilirubin in infected bile ducts
D. Increased bile salt secretion
E. Cholesterol supersaturation of bile

20. A 67-year-old woman with postprandial abdominal pain, fear of eating, and a 10-kg weight loss has severe atherosclerotic stenosis of the SMA and celiac trunk on CT angiography.

Which is the most appropriate treatment?

A. Long-term warfarin alone
B. Small frequent meals with nutritional support only
C. Proton pump inhibitor
D. Cholecystectomy
E. Endovascular angioplasty and stenting

21. A 19-year-old man is tackled during a football match and develops right upper quadrant pain. He also complains of pain at the tip of his right shoulder.

The shoulder pain is referred through which nerve?

A. Vagus nerve
B. Intercostal nerve
C. Phrenic nerve
D. Greater splanchnic nerve
E. Long thoracic nerve

22. A 58-year-old man with a history of injection drug use has palpable purpura on his legs, joint pain, and hematuria. Serum complement levels are low and cryoglobulins are present. Kidney biopsy shows membranoproliferative glomerulonephritis.

Which chronic infection is most likely responsible?

A. Hepatitis C
B. Hepatitis E
C. Hepatitis A
D. Epstein-Barr virus
E. Cytomegalovirus

23. A 52-year-old man has just completed 14 days of eradication therapy for H. pylori. He continues to take omeprazole daily and asks when he can be tested to confirm the infection is gone.

Which approach is most appropriate?

A. Perform a urea breath test tomorrow while continuing omeprazole at the same daily dose
B. Stop omeprazole 2 weeks before a urea breath test, done at least 4 weeks after ABX
C. Continue omeprazole, add bismuth for 4 weeks, then perform a urea breath test the next day
D. H. pylori IgG serology now
E. No testing is needed after a full course of therapy

24. Three weeks after nonoperative management of a liver laceration from a car crash, a 36-year-old woman has colicky RUQ pain, jaundice, and melena.

Which is the most likely diagnosis?

A. Biliary colic
B. Hemobilia
C. Esophageal varices
D. Peptic ulcer bleeding
E. Hepatic abscess

25. A 35-year-old man is admitted with full-thickness burns over 45% of his body surface area. On hospital day 3, he has an episode of coffee-ground emesis.

Which mechanism best explains this complication?

A. Urease-mediated mucosal injury
B. Inhibition of prostaglandin synthesis
C. Hypovolemia leading to gastric mucosal ischemia
D. Increased vagal stimulation of parietal cells
E. Gastrin-secreting tumor

26. A 55-year-old man with chronic GERD has endoscopy showing Barrett esophagus without dysplasia. He is started on a daily PPI.

Which is the most appropriate surveillance plan?

A. Esophagectomy
B. No further endoscopy is needed
C. Repeat endoscopy every 3–5 years
D. Repeat endoscopy every 3 months
E. Annual barium swallow

27. Several children at a childcare center develop fever, abdominal cramps, and small-volume bloody, mucoid stools. The outbreak spreads person to person despite no shared food source. Stool culture grows a non-motile, lactose-nonfermenting gram-negative rod that does not produce H2S.

Which organism is most likely responsible?

A. Enterotoxigenic E. coli
B. Bacillus cereus
C. Norovirus
D. Salmonella
E. Shigella

28. A 32-year-old woman meets diagnostic criteria for IBS. Her clinician explains that her abdominal pain partly reflects increased central amplification of signals from gut stretch receptors.

Which condition is she most likely to have as well?

A. Ulcerative colitis
B. Fibromyalgia
C. Primary biliary cholangitis
D. Celiac disease
E. Hemochromatosis

29. A 54-year-old woman with new ascites undergoes diagnostic paracentesis. Serum albumin is 3.0 g/dL and ascitic albumin is 1.1 g/dL. Ascitic neutrophils are 60/μL.

Which is the most likely cause of her ascites?

A. Tuberculous peritonitis
B. Peritoneal carcinomatosis
C. Nephrotic syndrome
D. Portal hypertension
E. Pancreatic ascites

30. A liver biopsy from a woman with acute liver failure in the third trimester of pregnancy is reviewed.

Which histologic finding is most characteristic of acute fatty liver of pregnancy?

A. Macrovesicular steatosis with Mallory-Denk bodies
B. PAS-positive globules in periportal hepatocytes
C. Noncaseating granulomas
D. Periportal fibrin deposits and hemorrhage
E. Microvesicular steatosis

31. A 31-year-old man has chronic diarrhea, bloating, and weight loss. He eats a regular diet that includes bread and pasta.

Which is the best initial test for the most likely diagnosis?

A. Hydrogen breath test
B. Upper endoscopy with duodenal biopsy
C. IgA anti-tTG antibodies with total serum IgA
D. Fecal elastase
E. Stool ova and parasites on three separate samples

32. A 58-year-old man had a cholecystectomy 6 years ago. He now has jaundice from a stone in the common bile duct. The stone is soft and brown, and bile culture grows E. coli.

Which best explains how this stone formed?

A. Calcium oxalate precipitation
B. Excess bile salt secretion
C. Cholesterol supersaturation in the gallbladder
D. Bacterial deconjugation of bilirubin within the bile duct
E. Chronic hemolysis

33. A 63-year-old woman with epigastric pain that worsens after meals undergoes endoscopy, which shows a 2-cm ulcer on the lesser curvature of the stomach.

Which is the most appropriate next step during this endoscopy?

A. Biopsy the ulcer margins
B. No biopsy; start a PPI only
C. Refer for partial gastrectomy
D. Inject the ulcer with adrenaline
E. Place a gastric stent

34. A 63-year-old obese woman with long-standing heartburn has a barium study showing the gastroesophageal junction and part of the stomach displaced above the diaphragm.

Which type of hernia is this?

A. Morgagni hernia
B. Sliding hiatal hernia
C. Umbilical hernia
D. Paraesophageal hiatal hernia
E. Bochdalek hernia

35. A 55-year-old woman with decompensated cirrhosis and tense ascites develops a rising creatinine and low urine output. Urine sodium is under 10 mmol/L, and creatinine does not improve after stopping diuretics and giving IV albumin. Kidney ultrasound is normal.

Which mechanism best explains her kidney failure?

A. Immune complex glomerulonephritis
B. Acute tubular necrosis from recent IV contrast exposure
C. Obstruction of the ureters by ascites
D. Splanchnic vasodilation causing renal vasoconstriction
E. Renal vein thrombosis

36. An abdominal x-ray of a 68-year-old woman with chronic gallstones shows a calcified gallbladder wall outlining the gallbladder.

This finding is associated with an increased risk of which condition?

A. Hepatocellular carcinoma
B. Acute pancreatitis
C. Primary biliary cholangitis
D. Gallbladder carcinoma
E. Hepatic angiosarcoma

37. A 74-year-old woman with A-fib who is not anticoagulated has sudden, severe periumbilical pain that began 3 hours ago. She is writhing in pain, but her abdomen is soft with only mild tenderness. Serum lactate is elevated.

Which is the most appropriate diagnostic test?

A. Colonoscopy with mucosal biopsy
B. Barium enema with air contrast
C. Plain abdominal x-ray
D. Abdominal ultrasound
E. CT angiography of the abdomen

38. A 29-year-old man with diarrhea-predominant IBS has urgent loose stools on workdays despite a low-FODMAP diet. He has no alarm features, and his workup is normal.

Which drug is most appropriate for his diarrhea?

A. Senna
B. Loperamide
C. Polyethylene glycol
D. Lactulose
E. Metoclopramide

39. A 26-year-old woman at 30 weeks' gestation develops fever, nausea, and jaundice 5 weeks after returning from a trip where she drank untreated water. Within days she becomes confused, with an INR of 2.8.

Which virus most likely caused her illness?

A. Hepatitis D virus
B. Hepatitis E virus
C. Hepatitis B virus
D. Hepatitis A virus
E. Hepatitis C virus

40. A 34-year-old woman has 3 weeks of right upper quadrant pain and increasing abdominal girth. She has polycythemia vera (positive JAK2 V617F mutation). She has tender hepatomegaly and shifting dullness. Jugular venous pressure is normal and there is no peripheral edema.

Which imaging study is most appropriate to evaluate the most likely diagnosis?

A. ERCP
B. Upper endoscopy
C. Doppler ultrasound of the hepatic veins
D. Transthoracic echocardiography of the right heart
E. HIDA scan

41. A 48-year-old woman has 3 liters per day of watery diarrhea that continues while she fasts. Bloods show hypokalemia and low gastric acid. The stool osmotic gap is 20 mOsm/kg.

Which is the most likely cause?

A. Lactose intolerance
B. Vipoma
C. Polyethylene glycol use
D. Ulcerative colitis
E. Exocrine pancreatic insufficiency

42. A 36-year-old aid worker returns after 8 months in the Caribbean with chronic diarrhea, weight loss, and megaloblastic anemia. Infectious stool studies are negative. Duodenal biopsy shows villous blunting. Celiac serology is negative.

Which is the most appropriate treatment?

A. Gluten-free diet
B. Pancreatic enzyme replacement
C. Lactose-free diet
D. Metronidazole for 5 days
E. Tetracycline and folic acid

43. A 55-year-old man with a history of duodenal ulcers presents with massive hematemesis and hypotension. Endoscopy shows a bleeding ulcer on the posterior wall of the duodenal bulb.

Which artery is most likely bleeding?

A. Short gastric arteries
B. Superior mesenteric artery
C. Splenic artery
D. Gastroduodenal artery
E. Left gastric artery

44. A 59-year-old woman recovered from ascending cholangitis after ERCP with removal of a common bile duct stone. Ultrasound shows more gallstones in her gallbladder.

Which is the most appropriate step to prevent recurrence?

A. Cholecystectomy
B. Repeat ERCP every 6 months
C. No further treatment
D. Long-term oral ABX
E. Ursodeoxycholic acid only

45. A 70-year-old man with A-fib develops acute mesenteric ischemia from an embolus.

Which artery is the embolus most likely to have occluded?

A. Splenic artery
B. Renal artery
C. Inferior mesenteric artery
D. Celiac trunk
E. Superior mesenteric artery

46. A 62-year-old obese man with 20 years of poorly controlled GERD undergoes endoscopy. Biopsies from salmon-colored mucosa above the GE junction show columnar epithelium with goblet cells.

This patient is at greatest risk of developing which of the following?

A. Adenocarcinoma of the distal esophagus
B. Squamous cell carcinoma of the upper esophagus
C. Gastric MALT lymphoma
D. Zenker diverticulum
E. Esophageal leiomyoma

47. A 31-year-old woman with mild-to-moderate ulcerative colitis limited to the left colon has achieved remission with a short course of corticosteroids.

Which drug is most appropriate for maintaining remission?

A. Ciprofloxacin
B. Long-term prednisone
C. Mesalazine (5-ASA)
D. Loperamide
E. Octreotide

48. A 30-year-old man has bloody diarrhea with tenesmus. Colonoscopy shows continuous inflammation from the rectum to the splenic flexure. His alkaline phosphatase is elevated, and MRCP shows multifocal strictures and dilations ("beading") of the intra- and extrahepatic bile ducts.

Besides colorectal cancer, this patient is at greatest risk for which malignancy?

A. Gastric MALT lymphoma
B. Cholangiocarcinoma
C. Small-bowel lymphoma
D. Hepatocellular carcinoma
E. Pancreatic adenocarcinoma

49. A 67-year-old man is resuscitated from cardiogenic shock after a large anterior myocardial infarction. Thirty-six hours later, AST is 4200 U/L, ALT is 3800 U/L and LDH is 5100 U/L. He has no history of liver disease, alcohol use, or paracetamol ingestion.

A liver biopsy would most likely show necrosis in which region?

A. Periportal hepatocytes (zone 1)
B. Kupffer cells only
C. Midzonal hepatocytes (zone 2)
D. Bile duct epithelium
E. Centrilobular hepatocytes (zone 3)

50. A 44-year-old man with heavy alcohol use has fever, jaundice, and tender hepatomegaly. Bilirubin is 14 mg/dL and his Maddrey discriminant function is 48. There is no infection or GI bleeding.

Which treatment is most appropriate in addition to abstinence and nutrition?

A. Prednisolone
B. N-acetylcysteine alone
C. Ursodeoxycholic acid
D. Liver transplant immediately
E. Phlebotomy

51. A 33-year-old woman with Budd-Chiari syndrome has worsening ascites despite 3 months of therapeutic anticoagulation. Venography shows a short, focal stenosis of the right hepatic vein. She has no cirrhosis or liver failure.

Which is the most appropriate next step?

A. Stop anticoagulation
B. Hepatic vein angioplasty with or without stenting
C. Immediate referral for liver transplant evaluation
D. Observation only
E. Start a beta-blocker

52. A 74-year-old man has sudden worsening of left lower quadrant pain, now diffuse, with a rigid abdomen, fever, and hypotension. CT shows free air and fecal contamination of the peritoneal cavity.

Which is the most appropriate management?

A. High-fiber diet
B. Resuscitation, IV ABX, and percutaneous drainage only
C. Outpatient oral ABX
D. IV ABX, resuscitation, and emergency laparotomy
E. Urgent colonoscopy

53. Several guests at a summer picnic develop abrupt nausea, vomiting, and abdominal cramps about 3 hours after eating potato salad. None have fever, and all recover within a day.

Which mechanism best explains the rapid onset?

A. Destruction of intestinal villi by viral replication
B. Ingestion of preformed heat-stable enterotoxin
C. Inhibition of the 60S ribosomal subunit
D. Invasion of colonic mucosa
E. In vivo production of a toxin that raises cAMP

54. A patient has a stone in the common bile duct and develops an infection of the obstructed bile.

Which combination of findings makes up Charcot triad?

A. Fever, right upper quadrant pain, jaundice
B. Painless jaundice, palpable gallbladder, weight loss
C. Right upper quadrant pain, Murphy sign, fever
D. Jaundice, pruritus, pale stools
E. Fever, hypotension, confusion

55. A 38-year-old woman with poorly controlled type 2 diabetes develops severe epigastric pain radiating to her back. Her blood sample looks milky, and lipase is markedly raised. She does not drink alcohol, and ultrasound shows no gallstones.

Which laboratory finding most likely explains her pancreatitis?

A. Elevated serum ferritin
B. Positive anti-mitochondrial antibody
C. Serum triglycerides of 2400 mg/dL
D. Serum calcium of 8.0 mg/dL
E. Elevated serum IgG4

56. A 47-year-old man with dyspepsia has a positive urea breath test. Endoscopy shows chronic gastritis.

Which region of the stomach is most typically affected first?

A. Cardia
B. Antrum
C. Body
D. Greater curvature only
E. Fundus

57. A 59-year-old woman being treated for diverticulitis has persistent fever and pain after 3 days of IV ABX. CT shows a 6-cm pericolic fluid collection with an enhancing rim and no free air.

Which is the most appropriate next step?

A. Colonoscopy
B. Continue IV ABX alone and repeat CT in 1 week
C. Increase dietary fiber
D. CT-guided percutaneous drainage plus ABX
E. Emergency colectomy with end colostomy

58. A 27-year-old man develops fever, nausea, dark urine, and jaundice 4 weeks after a backpacking trip through Central America, where he ate street food. ALT is 2100 U/L.

Which test best confirms the most likely diagnosis?

A. IgM anti-HAV
B. IgG anti-HAV
C. Heterophile antibody test
D. HBsAg
E. Anti-HCV antibody

59. A 47-year-old woman is admitted with mild gallstone pancreatitis. After 3 days her pain has resolved, she is eating, and her LFTs are normalizing. There is no cholangitis.

Which is the most appropriate next step?

A. Laparoscopic cholecystectomy before discharge
B. ERCP with sphincterotomy
C. Discharge with elective cholecystectomy in 6 months
D. No further intervention
E. Long-term ursodeoxycholic acid

60. A 50-year-old woman develops fever, eosinophilia, and severe hepatitis 1 week after her second general anesthetic in 3 months. Both used a halogenated inhaled anesthetic.

Which mechanism best explains her liver injury?

A. Copper accumulation
B. Immune reaction to drug-modified liver proteins
C. Hypoperfusion of the liver during repeated prolonged surgery
D. Bile duct injury during surgery
E. Excess NAPQI formation

61. A 52-year-old kidney transplant recipient on tacrolimus and mycophenolate develops odynophagia. Endoscopy shows several large, shallow, linear ulcers in the distal esophagus. Biopsy from the ulcer base shows enlarged cells with intranuclear and intracytoplasmic inclusions.

Which of the following is the most appropriate treatment?

A. Ganciclovir
B. Fluconazole
C. Acyclovir
D. Sucralfate
E. Prednisone

62. A 19-year-old man has a 6-month history of hand tremor, slurred speech, and new irritability. His AST and ALT are mildly elevated. Slit-lamp examination shows a golden-brown ring at the edge of each cornea. Serum ceruloplasmin is low.

A defect in which process best explains his condition?

A. Intestinal iron absorption
B. Secretion of alpha-1 antitrypsin from hepatocytes
C. Conjugation of bilirubin
D. Hepatic uptake of LDL cholesterol
E. Hepatocyte copper excretion into bile

63. An 80-year-old man with untreated gallstones has 2 days of repeated vomiting and epigastric fullness. CT shows air in the biliary tree, a distended stomach, and a large gallstone impacted in the duodenal bulb.

Which is the most likely diagnosis?

A. Gallstone pancreatitis
B. Bouveret syndrome
C. Gallstone ileus at the ileocecal valve
D. Acute cholangitis
E. Mirizzi syndrome

64. A 5-year-old boy had bloody diarrhea for 4 days after eating an undercooked hamburger. He is now pale with petechiae and decreased urine output. Bloods show hemoglobin 7.8 g/dL, platelets 42000/μL, creatinine 2.1 mg/dL and schistocytes on the blood smear.

Which intervention should be avoided?

A. Monitoring electrolytes and urine output
B. Careful IV fluid management
C. Red blood cell transfusion if symptomatic
D. ABX therapy
E. Dialysis if indicated

65. A 52-year-old man with diabetes, bronze skin, and cirrhosis develops dyspnea and peripheral edema. Echocardiography shows thickened ventricular walls with impaired diastolic filling and normal ejection fraction.

Which type of cardiomyopathy is most likely?

A. Hypertrophic obstructive cardiomyopathy
B. Peripartum cardiomyopathy
C. Arrhythmogenic right ventricular cardiomyopathy
D. Takotsubo cardiomyopathy
E. Restrictive cardiomyopathy from iron deposition

66. A 49-year-old woman with obesity and type 2 diabetes has steatosis on ultrasound and mildly raised ALT. She drinks rarely, and viral and autoimmune screens are negative.

Which is the most appropriate next step to assess her risk of advanced liver disease?

A. Serum ceruloplasmin
B. No further assessment is needed
C. Liver biopsy, since everyone with MASLD needs histologic staging
D. Calculate a FIB-4 score, followed by elastography if indicated
E. Repeat ultrasound in 5 years

67. A 58-year-old man with hypertension, type 2 diabetes, and a 30-pack-year smoking history reports 2 days of burning substernal chest pain that he attributes to "heartburn".

Which of the following is the most appropriate initial step?

A. Start a proton pump inhibitor
B. Test for H. pylori
C. Upper endoscopy
D. 12-lead ECG
E. Prescribe an antacid and reassess in 2 weeks

68. A 24-year-old woman has fatigue and jaundice. ALT is 940 U/L and AST 820 U/L, with mildly raised ALP. Viral serologies are negative. ANA and anti-smooth muscle antibody are positive, and serum IgG is elevated.

Which is the most appropriate treatment?

A. Tenofovir, with or without entecavir
B. Ursodeoxycholic acid, with or without cholestyramine
C. Cholestyramine
D. Therapeutic phlebotomy
E. Corticosteroids, with or without azathioprine

69. A 62-year-old man with cirrhosis is brought in confused and drowsy after 4 days of constipation. He has a flapping tremor when he holds his arms out. There is no fever or GI bleeding.

Which is the most appropriate first-line treatment?

A. Furosemide
B. Lorazepam
C. Haloperidol
D. Protein-free diet
E. Lactulose

70. A 28-year-old man with ulcerative colitis has been taking sulfasalazine for 2 years. He and his partner have been unable to conceive, and his semen analysis shows a low sperm count.

Which other adverse effect is associated with this drug?

A. Pulmonary fibrosis
B. Folate deficiency
C. Hypertrichosis
D. Ototoxicity
E. Hyperkalemia

71. A 60-year-old man with years of untreated duodenal ulcers has 2 weeks of early satiety and repeated vomiting of undigested food. A splashing sound is heard over the epigastrium when his abdomen is gently shaken.

Which acid-base disturbance is most likely?

A. Anion gap metabolic acidosis
B. Hyperchloremic, hyperkalemic metabolic acidosis
C. Respiratory acidosis
D. Non-anion gap metabolic acidosis
E. Hypochloremic, hypokalemic metabolic alkalosis

72. A 50-year-old woman has an isolated raised alkaline phosphatase on routine bloods. Her clinician orders a GGT.

What is the purpose of the GGT?

A. To assess renal function
B. To measure liver synthetic function alongside INR and albumin
C. To confirm that the ALP is from the liver rather than bone
D. To diagnose primary biliary cholangitis
E. To detect hemolysis

73. A 20-year-old woman has RLQ pain. When the examiner presses deeply on her left lower quadrant, she feels pain in the right lower quadrant.

What is this finding called?

A. Obturator sign
B. Cullen sign
C. Rovsing sign
D. Psoas sign
E. Murphy sign

74. A 38-year-old woman with long-standing ileal Crohn disease develops recurrent right upper quadrant pain after fatty meals. Ultrasound shows gallstones.

Which mechanism best explains her gallstones?

A. Increased gallbladder motility
B. Decreased ileal bile salt reabsorption, supersaturating bile with cholesterol
C. Autoimmune destruction of bile ducts
D. Increased bilirubin production from hemolysis, forming pigment stones in the gallbladder
E. Infection of the bile ducts

75. A 34-year-old man with HIV who stopped antiretroviral therapy a year ago presents with 2 weeks of painful swallowing. His CD4 count is 62/μL. Examination shows white plaques on the buccal mucosa that scrape off, leaving an erythematous base. He is hemodynamically stable and can tolerate oral fluids.

Which of the following is the most appropriate next step in management?

A. Oral acyclovir
B. IV ganciclovir
C. Empiric oral fluconazole
D. High-dose oral omeprazole
E. Upper endoscopy with biopsy before any treatment

76. A 39-year-old woman is diagnosed with achalasia on manometry after endoscopy excludes a mass. She is otherwise healthy and wants a durable treatment.

Which treatment is most appropriate?

A. Laparoscopic Nissen fundoplication
B. Metoclopramide
C. Esophageal stent
D. Pneumatic dilation of the LES
E. Long-term high-dose PPI

77. Six weeks after an episode of alcohol-related acute pancreatitis, a 45-year-old man has early satiety and epigastric fullness. CT shows a 7-cm well-defined fluid collection next to the pancreas. Its wall is made of fibrous and granulation tissue.

Which feature distinguishes this lesion from a true cyst?

A. It contains pancreatic enzymes
B. It can become infected
C. It is surrounded by a fibrous capsule
D. It has no epithelial lining
E. It is located next to the pancreas

78. A 57-year-old man with cirrhosis has a screening endoscopy showing large esophageal varices that have never bled. His heart rate is 88/min and blood pressure 124/78 mm Hg.

Which is the most appropriate treatment to prevent a first bleed?

A. Selective beta-blocker such as metoprolol or bisoprolol
B. Octreotide infusion
C. Long-term proton pump inhibitor
D. TIPS
E. Non-selective beta-blocker such as propranolol

79. A 50-year-old man has several gallstones found incidentally on an abdominal ultrasound done for kidney stones. He has never had abdominal pain, and his LFTs are normal.

Which is the most appropriate management of his gallstones?

A. Repeat ultrasound and LFTs every 6 months
B. ERCP with sphincterotomy
C. Elective laparoscopic cholecystectomy
D. No treatment; return if symptoms develop
E. Ursodeoxycholic acid for 2 years

Test C answer sheet

Name _____ Date _____ Score _____ / 79

1 A B C D E	**21** A B C D E	**41** A B C D E	**61** A B C D E
2 A B C D E	**22** A B C D E	**42** A B C D E	**62** A B C D E
3 A B C D E	**23** A B C D E	**43** A B C D E	**63** A B C D E
4 A B C D E	**24** A B C D E	**44** A B C D E	**64** A B C D E
5 A B C D E	**25** A B C D E	**45** A B C D E	**65** A B C D E
6 A B C D E	**26** A B C D E	**46** A B C D E	**66** A B C D E
7 A B C D E	**27** A B C D E	**47** A B C D E	**67** A B C D E
8 A B C D E	**28** A B C D E	**48** A B C D E	**68** A B C D E
9 A B C D E	**29** A B C D E	**49** A B C D E	**69** A B C D E
10 A B C D E	**30** A B C D E	**50** A B C D E	**70** A B C D E
11 A B C D E	**31** A B C D E	**51** A B C D E	**71** A B C D E
12 A B C D E	**32** A B C D E	**52** A B C D E	**72** A B C D E
13 A B C D E	**33** A B C D E	**53** A B C D E	**73** A B C D E
14 A B C D E	**34** A B C D E	**54** A B C D E	**74** A B C D E
15 A B C D E	**35** A B C D E	**55** A B C D E	**75** A B C D E
16 A B C D E	**36** A B C D E	**56** A B C D E	**76** A B C D E
17 A B C D E	**37** A B C D E	**57** A B C D E	**77** A B C D E
18 A B C D E	**38** A B C D E	**58** A B C D E	**78** A B C D E
19 A B C D E	**39** A B C D E	**59** A B C D E	**79** A B C D E
20 A B C D E	**40** A B C D E	**60** A B C D E	

Test A answers and explanations

1 **E**	2 **E**	3 **C**	4 **D**	5 **C**	6 **E**	7 **E**	8 **A**	9 **A**	10 **A**
11 **C**	12 **A**	13 **C**	14 **D**	15 **B**	16 **B**	17 **D**	18 **A**	19 **C**	20 **B**
21 **D**	22 **E**	23 **D**	24 **E**	25 **B**	26 **D**	27 **D**	28 **C**	29 **E**	30 **B**
31 **C**	32 **B**	33 **A**	34 **E**	35 **D**	36 **B**	37 **A**	38 **D**	39 **A**	40 **A**
41 **C**	42 **C**	43 **B**	44 **E**	45 **E**	46 **C**	47 **D**	48 **B**	49 **B**	50 **D**
51 **A**	52 **E**	53 **C**	54 **E**	55 **B**	56 **A**	57 **D**	58 **D**	59 **C**	60 **A**
61 **D**	62 **C**	63 **A**	64 **D**	65 **A**	66 **E**	67 **C**	68 **A**	69 **D**	70 **B**
71 **B**	72 **E**	73 **E**	74 **A**	75 **C**	76 **B**	77 **B**	78 **C**	79 **B**	80 **E**

1. E. Reassurance and a trial of a low-FODMAP diet

IBS · Step 2

IBS is a disorder of gut-brain interaction diagnosed clinically (Rome V criteria) once alarm features are absent and basic tests are normal. First-line management is reassurance, education, and a low-FODMAP diet, with symptom-directed drugs as needed.

D. Colonoscopy is not needed without alarm features in a patient under 45.

2. E. Squamous cell carcinoma of the esophagus

Esophageal motility disorders · Step 1

Long-standing achalasia raises the risk of esophageal squamous cell carcinoma, probably from chronic irritation by stagnant food. Aspiration pneumonia is the other key complication.

3. C. Alcoholic hepatitis

Iatrogenic and toxic liver injury · Step 2

Alcoholic hepatitis typically shows an AST:ALT ratio above 2:1 with AST rarely above 300 U/L, raised GGT, and macrocytosis. ALT stays relatively low partly because it needs pyridoxal phosphate (vitamin B6), which is often deficient in heavy drinkers.

D. Viral hepatitis usually causes ALT-predominant rises, often well above 1000 U/L.

E. MASLD typically has ALT higher than AST.

4. D. Anti-mitochondrial antibody

Autoimmune liver injury · Step 1

Primary biliary cholangitis typically affects middle-aged women with other autoimmune diseases (Hashimoto, scleroderma, rheumatoid arthritis, Sjögren). Pruritus, fatigue, xanthelasma, and a cholestatic pattern with non-dilated ducts are classic. More than 90% are AMA-positive.

B. Anti-smooth muscle antibody points to autoimmune hepatitis type 1, which has a hepatocellular (AST/ALT-predominant) pattern.

E. p-ANCA is associated with PSC, which typically affects men with ulcerative colitis.

5. C. Norovirus

Diarrhea · Step 1

Norovirus is highly contagious and causes outbreaks in closed settings such as cruise ships, with sudden vomiting and watery (non-inflammatory) diarrhea lasting 1–3 days.

6. E. Class B

Acute liver failure and cirrhosis · Step 2

Bilirubin 2–3 (2 points), albumin 2.8–3.5 (2 points), INR under 1.7 (1 point), mild ascites (2 points) and no encephalopathy (1 point) give 8 points: Class B (7–9). Creatinine is part of the MELD score, not Child-Pugh.

B. Creatinine is used in MELD, not Child-Pugh.

D. There are only three Child-Pugh classes: A, B, and C.

7. E. Prompt delivery

AFLP · Step 2

Hypoglycemia, coagulopathy, and encephalopathy in the third trimester point to acute fatty liver of pregnancy. The only cure is prompt delivery, with supportive care (glucose, blood products, ICU). The newborn should be tested for LCHAD deficiency.

C. Ursodeoxycholic acid treats intrahepatic cholestasis of pregnancy, which causes itching but not liver failure.

8. A. Increase dietary fiber

Diverticular diseases · Step 2

A high-fiber diet increases stool bulk and lowers intraluminal colonic pressure, the main driver of diverticula. Avoiding nuts and seeds is no longer recommended.

9. A. Cryptosporidium

Diarrhea · Step 1

Cryptosporidium causes chronic, severe watery diarrhea in immunocompromised patients, especially with low CD4 counts. Oocysts are acid-fast on stool staining. Restoring immunity with antiretroviral therapy is key.

10. A. False diverticula of mucosa and submucosa herniating at vasa recta entry points

Diverticular diseases · Step 1

Colonic diverticula are false (pulsion) diverticula. Mucosa and submucosa herniate through weak points in the muscularis externa where the vasa recta enter. High intraluminal pressure in the sigmoid, from low fiber, constipation, and aging, drives their formation.

D. An outpouching above the cricopharyngeus is a Zenker diverticulum.

E. A vitelline duct remnant is a Meckel diverticulum, which is a true diverticulum.

11. C. Magnesium hydroxide

PUD · Step 1

Magnesium hydroxide is an osmotic agent and causes diarrhea, while aluminum hydroxide causes constipation (combination products balance the two). Calcium carbonate can cause hypercalcemia and rebound acid secretion.

12. A. Herpes simplex virus

Esophagitis · Step 1

HSV esophagitis causes small, punched-out ("volcano") ulcers, with multinucleated giant cells and Cowdry A inclusions in squamous cells at the ulcer edge. It is treated with acyclovir.

C. Candida causes white plaques, not punched-out ulcers.

E. CMV causes large, shallow, linear ulcers, with inclusions in endothelial and stromal cells at the ulcer base.

13. C. CT AP with contrast

Diverticular diseases · Step 2

CT confirms diverticulitis and identifies complications such as abscess or perforation. Colonoscopy and barium enema are avoided during acute inflammation because of perforation risk. Colonoscopy is done 6–8 weeks after recovery to exclude malignancy.

D. Colonoscopy is contraindicated in acute diverticulitis; it is deferred until after recovery.

E. Barium enema risks perforation and barium peritonitis.

14. D. Calcium oxalate

IBD · Step 1

With ileal disease or resection, bile salt and fat malabsorption leave fat in the lumen, where it binds calcium. Oxalate that would normally bind calcium is left free, is absorbed in the colon, and forms calcium oxalate stones (enteric hyperoxaluria). Loss of bile salt recycling also predisposes to cholesterol gallstones.

15. B. Cardiovascular disease

Metabolic liver injury · Step 2

This is MASLD (formerly NAFLD), driven by metabolic syndrome. Although it can progress to MASH, cirrhosis, and HCC, cardiovascular disease is the leading cause of death. MASLD also raises the risk of type 2 diabetes, chronic kidney disease, and extrahepatic cancers.

16. B. Increased vagal (CN X) stimulation of parietal cells

Gastritis · Step 1

Cushing ulcers follow brain injury. Raised ICP increases vagal output, and acetylcholine stimulates parietal cells to secrete more HCl. Like Curling ulcers, these are erosions that do not penetrate the muscularis mucosae.

17. D. Surveillance colonoscopy every 1–3 years starting now

IBD · Step 2

Colorectal cancer risk in ulcerative colitis rises with duration and extent of disease. Surveillance colonoscopy starts about 8 years after diagnosis (or at diagnosis if PSC is present) and is repeated every 1–3 years.

18. A. Hepatic adenoma

Liver tumors · Step 1

Hepatic adenomas are linked to oral contraceptives and anabolic steroids, and larger ones (over 5 cm) can rupture and bleed. Management includes stopping estrogen and resecting large or bleeding adenomas.

C. Focal nodular hyperplasia has a central stellate scar and rarely bleeds.

19. C. CT abdomen with IV contrast

Traumatic liver injury · Step 2

A hemodynamically stable patient with suspected blunt abdominal trauma gets a contrast CT to identify and grade solid-organ injury. Most stable liver injuries are managed without surgery, with angioembolization if there is active bleeding.

E. Emergency laparotomy is for unstable patients with a positive FAST or peritonitis.

20. B. Atherosclerotic mesenteric arteries that cannot meet post-meal oxygen demand

Intestinal ischemia · Step 1

Chronic mesenteric ischemia ("intestinal angina") results from atherosclerosis of at least 2 of the celiac trunk, SMA, and IMA. Blood flow is adequate at rest but cannot meet the higher demand after eating, causing postprandial pain, fear of eating, and weight loss. CT angiography confirms the diagnosis.

21. D. Water-soluble contrast esophagram

Esophagitis · Step 2

Chest pain, crepitus, and pneumomediastinum suggest esophageal perforation. A contrast esophagram with water-soluble contrast (with CXR or CT) confirms the leak; treatment is usually surgical, with ABX.

B. Endoscopy can enlarge a perforation.

E. Barium leaking into the mediastinum causes severe mediastinitis, so water-soluble contrast is used first.

22. E. The gallbladder does not fill with tracer

Cholecystitis · Step 2

In acute cholecystitis, the cystic duct is blocked, so the radiotracer passes into the bile ducts and duodenum but never fills the gallbladder.

A. Tracer not reaching the duodenum suggests common bile duct obstruction.

23. D. Portal vein

Vascular liver injury · Step 1

The liver has a dual blood supply: the portal vein provides about 75% of hepatic blood flow, and the hepatic artery about 25% (but roughly half the oxygen). This dual supply makes hepatic infarction uncommon.

24. E. Ileocecal valve

Cholecystitis · Step 1

Gallstone ileus: a large stone erodes through a cholecystoenteric fistula (usually cholecystoduodenal) and obstructs the narrowest part of the small bowel, the ileocecal valve. Rigler triad on imaging is pneumobilia, small bowel obstruction, and an ectopic gallstone.

B. A stone obstructing the pylorus or duodenum causes Bouveret syndrome (gastric outlet obstruction), not distal small bowel obstruction.

25. B. Falling IgA anti-tTG levels

Malabsorption · Step 2

Treatment is a lifelong gluten-free diet (avoiding wheat, rye, and barley). Celiac antibodies such as IgA anti-tTG fall with good adherence, and persistently raised levels suggest ongoing gluten exposure.

A. HLA type does not change; HLA-DQ2/DQ8 testing helps rule celiac disease out, not monitor it.

26. D. Descemet membrane

Metabolic liver injury · Step 1

Kayser-Fleischer rings are copper deposits in Descemet membrane at the corneal periphery. They are seen in nearly all patients with neurologic Wilson disease.

A. Copper in the lens capsule causes a sunflower cataract, a separate finding.

27. D. Raised central venous pressure transmitted to the hepatic veins

Vascular liver injury · Step 1

Right heart failure raises central venous pressure, which backs up into the hepatic veins and sinusoids. Zone 3 congestion and atrophy produce the mottled "nutmeg liver". Chronic congestion can lead to cardiac cirrhosis.

A. Portal vein thrombosis causes portal hypertension without centrilobular congestion.

C. Budd-Chiari syndrome also causes a nutmeg pattern, but JVD and hepatojugular reflux point to a cardiac cause.

28. C. Increased estrogen from reduced hepatic clearance

Acute liver failure and cirrhosis · Step 1

The cirrhotic liver clears estrogen poorly, and more androgen is converted to estrogen peripherally. The excess estrogen causes gynecomastia, testicular atrophy, palmar erythema, and spider angiomas.

29. E. IV heparin and emergency revascularization

Intestinal ischemia · Step 2

Acute mesenteric ischemia needs immediate anticoagulation with heparin, fluid resuscitation, and urgent revascularization (embolectomy or endovascular therapy). Laparotomy is used to assess bowel viability and resect necrotic segments.

30. B. Emergency surgical repair

PUD · Step 2

Free air with peritonitis means a perforated ulcer, which is a surgical emergency. Resuscitate with IV fluids, IV PPI, and broad-spectrum ABX, then proceed to surgery.

A. Barium leaking into the peritoneum causes severe peritonitis; water-soluble contrast is used if contrast is needed at all.

C. Endoscopy insufflates air and can worsen a known perforation.

31. C. Retroperitoneal hemorrhage from necrotizing pancreatitis

Acute pancreatitis · Step 1

In necrotizing pancreatitis, enzymes erode blood vessels and blood tracks through the retroperitoneum. Flank bruising is Grey Turner sign and periumbilical bruising is Cullen sign. Both suggest severe disease.

32. B. Decreased LES tone and increased intragastric pressure

GERD · Step 1

Progesterone relaxes smooth muscle and lowers LES tone, and the enlarging uterus raises intragastric pressure. Obesity and gastroparesis also raise intragastric pressure. Smoking, alcohol, and caffeine lower LES tone.

33. A. Venous congestion reducing oxygen delivery to the bowel

Intestinal ischemia · Step 1

Mesenteric venous thrombosis occurs with hypercoagulable states (factor V Leiden, estrogen, malignancy) or portal hypertension. Blocked outflow causes venous congestion and bowel wall edema, which reduces oxygen delivery. It is treated with anticoagulation.

34. E. Laparoscopic Nissen fundoplication

GERD · Step 2

Refractory, confirmed GERD is treated surgically with Nissen fundoplication: the gastric fundus is wrapped around the lower esophagus to reinforce the LES. Normal peristalsis should be confirmed first.

A. Botulinum toxin lowers LES tone; it is used for achalasia.

B. Pneumatic dilation is an achalasia treatment.

C. Heller myotomy cuts the LES. It treats achalasia and would worsen reflux.

35. D. Enteropathy-associated T-cell lymphoma

Malabsorption · Step 1

Poorly controlled celiac disease raises the risk of enteropathy-associated T-cell lymphoma of the small bowel, as well as small bowel adenocarcinoma. A strict gluten-free diet lowers this risk.

36. B. Appendiceal neuroendocrine (carcinoid) tumor

Appendicitis · Step 1

Neuroendocrine (carcinoid) tumors are the most common appendiceal neoplasm, often found incidentally at the tip of the appendix. Small tumors (under 2 cm) are usually cured by appendectomy.

37. A. Chronic infection with high replication and infectivity

Infectious liver injury · Step 1

HBsAg for more than 6 months means chronic infection. Positive HBeAg and high HBV DNA show active viral replication and high infectivity, so her baby needs HBV vaccine and HBIG at birth.

B. Low replication would show negative HBeAg, positive anti-HBe, and low HBV DNA.

38. D. Secretin stimulation test

PUD · Step 1

Zollinger-Ellison syndrome is caused by a gastrin-secreting tumor (gastrinoma), associated with MEN1. Secretin normally inhibits gastrin release but paradoxically increases it from gastrinoma cells. Ulcers beyond the duodenum and diarrhea from acid inactivating pancreatic enzymes are clues.

C. Urinary 5-HIAA is used to detect carcinoid tumors.

39. A. Loperamide

IBD · Step 2

This is toxic megacolon. Antimotility agents (loperamide) and opioids worsen colonic dilation and can precipitate perforation. Management is bowel rest, IV fluids, IV steroids, ABX, and early surgical consultation. VTE prophylaxis is still needed because IBD is hypercoagulable.

40. A. Upper endoscopy

Esophageal motility disorders · Step 2

Older age, a short history, and marked weight loss suggest pseudoachalasia from a tumor at the gastroesophageal junction. Endoscopy must exclude malignancy before achalasia is diagnosed or treated.

41. C. Blood glucose

Acute liver failure and cirrhosis · Step 1

Hypoglycemia is one of the "big 4" complications of acute liver failure, alongside encephalopathy, coagulopathy, and infection. The failing liver cannot make glucose or release it from glycogen. A sudden drop in consciousness needs an immediate glucose check.

42. C. Corticosteroids

Autoimmune liver injury · Step 2

IgG4-related sclerosing cholangitis often occurs with autoimmune pancreatitis and can mimic PSC or cancer. Unlike PSC, it responds well to corticosteroids.

43. B. Low pH, low HCO_3^-, normal anion gap, low $PaCO_2$

Diarrhea · Step 1

Diarrhea loses bicarbonate-rich intestinal fluid, causing a non-anion-gap (hyperchloremic) metabolic acidosis with respiratory compensation (low $PaCO_2$). Hypokalemia and hypovolemia often accompany it.

E. High pH with high bicarbonate is metabolic alkalosis, typical of vomiting.

44. E. Normal urinary D-xylose; low fecal elastase

Malabsorption · Step 1

D-xylose is absorbed without pancreatic enzymes, so urinary excretion is normal in maldigestion (pancreatic insufficiency) and low in mucosal malabsorption such as celiac disease. Low fecal elastase confirms exocrine pancreatic insufficiency, which is treated with enzyme replacement.

45. E. Goal-directed IV fluids such as Hartmann solution

Acute pancreatitis · Step 2

Third spacing causes hypovolemia in pancreatitis. Early goal-directed IV fluids (Hartmann solution is commonly used), pain control and early feeding are the core of treatment. ABX are only for proven infection, and enteral feeding is preferred over TPN.

B. Prophylactic ABX do not improve outcomes; they are reserved for infected necrosis.

46. C. Gastric carcinoid (neuroendocrine) tumor

Gastritis · Step 1

Low acid removes feedback on G cells, and the resulting high gastrin stimulates enterochromaffin-like (ECL) cells to proliferate, which can lead to gastric carcinoid tumors. Autoimmune gastritis also raises the risk of gastric adenocarcinoma.

47. D. Body and fundus

Gastritis · Step 1

Autoimmune gastritis targets parietal cells, which sit in the body and fundus. Their loss reduces HCl and intrinsic factor, leading to B12 deficiency (pernicious anemia, megaloblastic anemia, neuropathy). It causes fundal atrophy rather than erosions. H. pylori gastritis typically involves the antrum.

B. The antrum is the classic site of H. pylori gastritis and contains G cells, not parietal cells.

48. B. High antigenic variation of its envelope glycoproteins

Infectious liver injury · Step 1

HCV's RNA polymerase has no proofreading, so its envelope glycoproteins vary constantly (hypervariable regions). This lets it escape antibodies, explains frequent chronic infection, and makes vaccine design difficult.

C. Needing another virus to replicate describes HDV, which depends on HBV.

49. B. Replenishing glutathione to neutralize NAPQI

Iatrogenic and toxic liver injury · Step 1

In overdose, glucuronidation and sulfation are saturated, so more paracetamol is converted by CYP2E1 to the toxic metabolite NAPQI. Once glutathione is used up, NAPQI damages hepatocytes (zone 3 necrosis). N-acetylcysteine restores glutathione and is most effective within 8 hours.

D. Binding the drug in the gut describes activated charcoal, which helps only within 1–2 hours.

50. D. Toxin-mediated ADP-ribosylation of Gs, increasing cAMP

Diarrhea · Step 1

Cholera toxin permanently activates Gs by ADP-ribosylation, raising cAMP. This opens CFTR chloride channels and drives large-volume secretory diarrhea without mucosal damage. Treatment is aggressive oral or IV rehydration.

B. ETEC heat-stable toxin activates guanylate cyclase; cholera toxin acts through cAMP.

51. A. Cholecystoduodenal

Biliary fistulas · Step 1

Air in the biliary tree without prior instrumentation suggests a biliary-enteric fistula. Cholecystoduodenal fistulas are the most common, because the gallbladder lies right next to the duodenum, and they are the usual route for gallstone ileus.

52. E. Anti-centromere antibody

Autoimmune liver injury · Step 1

PBC is associated with other autoimmune diseases, including limited systemic sclerosis (CREST: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, telangiectasia), which is linked to anti-centromere antibodies.

C. Anti-Scl-70 is associated with diffuse systemic sclerosis.

53. C. Osmotic laxative such as polyethylene glycol

IBS · Step 2

Laxatives treat constipation in IBS-C; polyethylene glycol is a common first choice. Secretagogues such as linaclotide are options if laxatives fail.

B. Loperamide and opioids slow transit and would worsen constipation.

54. E. Decreased LES pressure; decreased distal peristalsis

Esophageal motility disorders · Step 1

In scleroderma (CREST), the smooth muscle of the distal esophagus atrophies and is replaced by fibrous tissue. LES tone falls, causing severe reflux, and distal peristalsis weakens. Treatment centers on PPIs. Achalasia is the opposite: high LES tone.

C. Simultaneous, uncoordinated contractions describe diffuse esophageal spasm.

D. This pattern describes achalasia.

55. B. Azathioprine

Acute pancreatitis · Step 1

Drugs are the "D" in I GET SMASHED. Azathioprine is a classic cause of acute pancreatitis, usually within the first weeks of therapy. Others include valproate, thiazides, furosemide, sulfonamides, and didanosine.

56. A. String sign; Crohn disease

IBD · Step 1

The "string sign" is a narrowed terminal ileum from transmural inflammation and fibrosis in Crohn disease. "Lead pipe" colon (loss of haustra) is typical of chronic ulcerative colitis.

57. D. Serum ceruloplasmin and 24-hour urine copper

Metabolic liver injury · Step 2

Wilson disease is screened with serum ceruloplasmin (low) and 24-hour urine copper (high), plus a slit-lamp examination for Kayser-Fleischer rings. Liver copper or ATP7B testing confirms unclear cases.

58. D. Pancreatic enzyme replacement with meals

Malabsorption · Step 2

Cystic fibrosis causes exocrine pancreatic insufficiency (maldigestion), shown by low fecal elastase. Pancreatic enzyme replacement with meals, plus fat-soluble vitamin supplements, treats the steatorrhea.

59. C. IV PPI and upper endoscopy within 24 hours

PUD · Step 2

For upper GI bleeding from suspected peptic ulcer disease, resuscitate, start an IV PPI, and perform endoscopy (EGD) within 24 hours to diagnose and treat the bleeding point. Stop the NSAID and test for H. pylori.

D. Surgery is reserved for bleeding that endoscopy (and angiographic embolization) cannot control.

60. A. MRCP

Autoimmune liver injury · Step 2

Primary sclerosing cholangitis affects intra- and extrahepatic bile ducts, so it is diagnosed by cholangiography: MRCP shows multifocal strictures and dilations ("beading"). Liver biopsy is reserved for suspected small-duct PSC with a normal MRCP.

D. A HIDA scan evaluates acute cholecystitis.

61. D. ERCP with stone extraction

Choledocholithiasis · Step 2

Bilirubin over 4 mg/dL with a dilated CBD gives a high likelihood of choledocholithiasis, so proceed directly to ERCP to remove the stone, followed by cholecystectomy during the same admission. MRCP or endoscopic ultrasound is used when the likelihood is intermediate.

A. Cholecystectomy alone leaves the CBD stone in place.

62. C. Pancreatic head or other periampullary malignancy

Biliary malignancies · Step 2

Courvoisier law: painless jaundice with a palpable, nontender gallbladder is unlikely to be caused by gallstones. A slowly growing tumor below the cystic duct lets a healthy gallbladder distend. Gallstones usually scar the gallbladder, so it cannot enlarge, and they cause pain.

A. CBD stones usually cause painful jaundice, and the gallbladder is typically shrunken from chronic inflammation.

63. A. Hepatitis B vaccine plus HBIG within 12 hours of birth

Infectious liver injury · Step 2

Perinatal HBV transmission is highly likely to cause chronic infection (about 90%). Giving both the first vaccine dose and HBIG within 12 hours of birth prevents most infections.

64. D. Oral vancomycin

Diarrhea · Step 2

For an initial non-fulminant episode of C. difficile infection, oral fidaxomicin or oral vancomycin is first-line. Stop the inciting ABX if possible.

A. Antimotility agents are avoided because they may precipitate toxic megacolon.

E. IV vancomycin does not reach the colonic lumen, so it is ineffective for C. difficile.

65. A. Misfolded protein accumulating in the hepatocyte endoplasmic reticulum

Metabolic liver injury · Step 1

In alpha-1 antitrypsin deficiency (PiZZ), the misfolded Z protein polymerizes and gets trapped in the hepatocyte ER, which is toxic to the liver (gain of function). Too little A1AT reaches the blood, so neutrophil elastase destroys alveolar walls, causing lower-lobe panacinar emphysema (loss of function).

D. Unopposed elastase explains the lung disease, not the liver disease.

66. E. Add a daily proton pump inhibitor

Gastritis · Step 2

For patients at high risk of NSAID-induced ulcers who must continue the drug, adding a PPI (or switching to a COX-2 selective inhibitor) reduces GI injury. Misoprostol is an alternative but often causes diarrhea.

A. Corticosteroids combined with NSAIDs increase ulcer risk.

C. Aspirin adds to GI bleeding risk.

67. C. It increases cholesterol secretion into bile

Cholelithiasis · Step 1

Estrogen increases hepatic cholesterol uptake and synthesis (upregulating HMG-CoA reductase), so more cholesterol enters bile. Progesterone in pregnancy also slows gallbladder emptying. Both help explain the "female" and "fertile" Fs.

68. A. Outpatient clear liquid diet and close follow-up

Diverticular diseases · Step 2

Uncomplicated diverticulitis in a stable patient who can take oral fluids can be managed as an outpatient with diet modification, pain relief, and close follow-up. Oral ABX (e.g. amoxicillin-clavulanate) are given selectively.

B. Colonoscopy is avoided during acute inflammation and is done 6–8 weeks after recovery.

69. D. Reassurance; no biopsy or treatment needed

Liver tumors · Step 2

Peripheral nodular enhancement with centripetal fill-in is typical of a hemangioma, the most common benign liver tumor. Asymptomatic hemangiomas need no treatment, and biopsy is avoided because of bleeding risk.

70. B. Window period

Infectious liver injury · Step 1

In the window period HBsAg has cleared but anti-HBs has not yet appeared, so IgM anti-HBc is the only reliable marker of infection. Anti-HBe may be positive. Once anti-HBs appears and anti-HBc switches to IgG, he has recovered.

C. Chronic infection requires HBsAg to be positive for at least 6 months.

D. Recovery requires anti-HBs, and anti-HBc would be IgG.

71. B. Inducing emesis

Esophagitis · Step 2

After a caustic (acid or alkali) ingestion, never induce vomiting or attempt neutralization: re-exposing the esophagus to the agent worsens injury. Manage the airway, keep the patient NPO, and grade the injury with early endoscopy. Perforation requires surgery.

72. E. Sodium restriction and spironolactone, with furosemide added as needed

Acute liver failure and cirrhosis · Step 2

First-line ascites treatment is dietary sodium restriction plus spironolactone (blocking secondary hyperaldosteronism), often with furosemide. Large-volume paracentesis with albumin, TIPS, or transplant are used for tense or refractory ascites.

73. E. Anticoagulation and stopping the oral contraceptive

Vascular liver injury · Step 2

All patients with Budd-Chiari syndrome are anticoagulated, and the underlying cause is treated (here, stopping estrogen). Treatment then steps up if needed: angioplasty or stenting for short venous narrowing, then TIPS, then liver transplant for fulminant failure or cirrhosis.

74. A. Blocking H2 receptors, lowering cAMP in parietal cells

GERD · Step 1

H2-receptor antagonists competitively block histamine H2 receptors on parietal cells, reducing cAMP and acid secretion. They are less potent than PPIs and suit milder symptoms.

75. C. Take the tablet with a full glass of water and stay upright for at least 30 minutes

Esophagitis · Step 1

Bisphosphonates, doxycycline, KCl, and NSAIDs cause pill-induced esophagitis when the tablet lodges in the esophagus and injures the mucosa directly. A full glass of water and staying upright clear the pill into the stomach. Treatment is stopping or changing the offending medication.

A. Food (and calcium or milk) markedly reduces bisphosphonate absorption; it should be taken fasting.

D. Milk contains calcium, which binds alendronate.

E. Calcium binds alendronate and blocks its absorption.

76. B. Enterolithotomy to remove the stone from the bowel

Biliary fistulas · Step 2

Gallstone ileus is treated by enterolithotomy (opening the bowel to remove the stone), which relieves the obstruction. Cholecystectomy and fistula repair are usually deferred, especially in frail patients, because a combined procedure carries higher risk.

77. B. Graded-compression ultrasound, followed by MRI if inconclusive

Appendicitis · Step 2

In pregnancy, ultrasound is used first to avoid radiation, then MRI without gadolinium if ultrasound is inconclusive. The appendix may be displaced upward by the uterus, so pain can be atypical. Delaying treatment risks perforation, which threatens the fetus.

78. C. It is an HDV superinfection and is likely to become chronic

Infectious liver injury · Step 1

HDV needs HBsAg to form its envelope, so it only infects people with HBV. A negative IgM anti-HBc shows the HBV is chronic, so this is a superinfection, which is usually more severe and becomes chronic. In coinfection, both are acquired together and IgM anti-HBc is positive.

A. Coinfection would show a positive IgM anti-HBc, because the HBV infection would be acute.

B. There is no hepatitis C vaccine; HDV is prevented by the hepatitis B vaccine.

79. B. Escherichia coli

Ascending cholangitis · Step 1

Ascending cholangitis is caused by gut organisms climbing into obstructed bile, most often E. coli, followed by Klebsiella, Enterococcus, and anaerobes. Empiric ABX (e.g. piperacillin-tazobactam) cover these.

80. E. Sucralfate

PUD · Step 1

Sucralfate polymerizes in an acidic environment into a sticky gel that binds to the ulcer base and protects it. Because it needs acid to activate, it works poorly with acid-suppressing drugs.

B. Misoprostol is a PGE1 analog that increases mucus and bicarbonate secretion.

Test B answers and explanations

1 **D**	2 **B**	3 **B**	4 **B**	5 **B**	6 **D**	7 **B**	8 **D**	9 **C**	10 **B**
11 **C**	12 **C**	13 **D**	14 **D**	15 **B**	16 **C**	17 **B**	18 **C**	19 **E**	20 **A**
21 **C**	22 **B**	23 **A**	24 **D**	25 **E**	26 **E**	27 **C**	28 **C**	29 **A**	30 **B**
31 **A**	32 **C**	33 **A**	34 **A**	35 **A**	36 **E**	37 **B**	38 **D**	39 **E**	40 **A**
41 **E**	42 **E**	43 **C**	44 **B**	45 **B**	46 **E**	47 **D**	48 **C**	49 **A**	50 **C**
51 **D**	52 **D**	53 **E**	54 **D**	55 **E**	56 **E**	57 **E**	58 **A**	59 **C**	60 **A**
61 **B**	62 **E**	63 **D**	64 **B**	65 **D**	66 **A**	67 **B**	68 **D**	69 **A**	70 **E**
71 **E**	72 **A**	73 **C**	74 **A**	75 **C**	76 **A**	77 **D**	78 **C**	79 **D**	

1. D. HCV RNA by PCR

Infectious liver injury · Step 2

A positive anti-HCV antibody shows exposure but not whether the infection is current; the antibody stays positive for life, even after cure. HCV RNA confirms active infection. If positive, assess fibrosis (e.g. APRI, FIB-4, or elastography) before treatment.

C. Anti-HCV antibodies are not protective; reinfection is possible.

2. B. Proton pump inhibitor

Esophageal motility disorders · Step 2

Esophageal involvement in scleroderma lowers LES tone and causes severe reflux, so PPIs are the mainstay of treatment.

C. Calcium channel blockers (often used for Raynaud phenomenon) lower LES tone further and can worsen reflux.

E. Dilation, botulinum toxin, and myotomy lower LES pressure and are used in achalasia.

3. B. Reverse transcription of pregenomic RNA into DNA

Infectious liver injury · Step 1

Although HBV is a DNA virus, it replicates through an RNA intermediate. Host RNA polymerase II transcribes pregenomic RNA from cccDNA, and the viral reverse transcriptase copies it back into pdsDNA. Tenofovir and entecavir inhibit this reverse transcriptase.

A. Polyprotein cleavage is an +ssRNA virus step (e.g. HCV NS3/4A protease inhibitors).

D. No current drug removes cccDNA, which is why HBV is rarely cured.

4. B. Diverticulosis

Diverticular diseases · Step 1

Diverticulosis is the most common cause of major lower GI bleeding in older adults. It is typically painless, sudden, and large in volume because a vasa recta artery ruptures. Most bleeds stop on their own; colonoscopy finds and treats the source.

A. An anal fissure causes painful, small-volume bleeding.

C. Ischemic colitis causes pain with bleeding.

5. B. Iron-deficiency anemia

IBS · Step 2

Alarm features include iron-deficiency anemia, rectal bleeding, unintentional weight loss, nocturnal symptoms, onset after age 50, and a family history of colorectal cancer or IBD. They call for investigation such as colonoscopy. The other options are typical of IBS.

6. D. Long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD)

AFLP · Step 1

AFLP is linked to fetal LCHAD deficiency, a defect in mitochondrial beta-oxidation of long-chain fatty acids. Fatty acid metabolites from the fetus overwhelm the heterozygous mother's liver, causing microvesicular steatosis.

7. B. Gastric adenocarcinoma

PUD · Step 1

CagA-positive H. pylori strains cause more severe inflammation and carry a higher risk of peptic ulcers and gastric adenocarcinoma. VacA is another cytotoxin associated with more severe disease.

8. D. Noncaseating granulomas with transmural inflammation

IBD · Step 1

Crohn disease shows skip lesions, transmural inflammation (causing fistulas and strictures), noncaseating granulomas, and Th1-mediated inflammation. It most often involves the terminal ileum and spares the rectum. Smoking worsens Crohn disease.

B. Caseating granulomas suggest intestinal tuberculosis.

C. Inflammation limited to the mucosa and submucosa, with crypt abscesses, is typical of ulcerative colitis.

9. C. Campylobacter jejuni

Diarrhea · Step 1

Campylobacter jejuni causes inflammatory diarrhea from undercooked poultry and is the classic infectious trigger of Guillain-Barré syndrome, through molecular mimicry between bacterial and nerve gangliosides.

10. B. Calcium binding to free fatty acids (saponification)

Acute pancreatitis · Step 1

Pancreatic lipase digests surrounding fat, releasing fatty acids that combine with calcium (saponification). This lowers serum calcium, causing tingling and tetany. Low calcium is also a marker of more severe disease.

11. C. Pancreaticoduodenectomy (Whipple procedure)

Biliary malignancies · Step 2

A resectable tumor in the pancreatic head or periampullary region (the "double duct sign" shows both ducts blocked) is treated with pancreaticoduodenectomy (Whipple procedure), usually with chemotherapy. Stenting relieves jaundice but does not treat the cancer.

D. Distal pancreatectomy is used for tumors of the pancreatic body and tail.

12. C. Urgent ERCP for biliary drainage

Ascending cholangitis · Step 2

Charcot triad (fever, RUQ pain, jaundice) plus hypotension and confusion is Reynolds pentad: severe ascending cholangitis with septic shock. ABX alone are not enough; the infected, obstructed bile must be drained urgently, usually by ERCP.

13. D. Hypoglycemia with prolonged INR

AFLP · Step 2

AFLP causes true liver failure, so hypoglycemia and coagulopathy (raised INR, low fibrinogen) are common. HELLP is defined by hemolysis, raised liver enzymes, and low platelets, usually with hypertension, and rarely causes hypoglycemia. Both are treated with delivery.

14. D. Polyarteritis nodosa

Infectious liver injury · Step 1

Polyarteritis nodosa is a medium-vessel vasculitis classically associated with hepatitis B. Immune complexes damage medium arteries, causing microaneurysms, renal hypertension, mononeuritis multiplex, and skin nodules. Lungs are spared.

A. Mixed cryoglobulinemia is mainly associated with hepatitis C.

15. B. Start empiric IV acyclovir

Infectious liver injury · Step 2

HSV hepatitis is rare but can be fulminant, especially in the third trimester or in immunocompromised patients. It causes very high aminotransferases, often with little jaundice. IV acyclovir should be started empirically while HSV PCR results are pending.

16. C. Colovesical fistula

Diverticular diseases · Step 2

Pneumaturia, fecaluria, and recurrent polymicrobial UTIs suggest a colovesical (enterovesical) fistula, most often from diverticulitis. CT is the usual first test (air in the bladder), and treatment is surgical.

17. B. Vinyl chloride

Liver tumors · Step 1

Hepatic angiosarcoma, a malignant vascular tumor, is linked to vinyl chloride (used to make PVC plastics), arsenic and Thorotrast.

A. Aflatoxin is linked to hepatocellular carcinoma.

C. Aniline dyes are linked to bladder cancer.

18. C. Loss of inhibitory nitric oxide- and VIP-producing neurons in the myenteric plexus

Esophageal motility disorders · Step 1

Achalasia results from loss of inhibitory neurons in the myenteric (Auerbach) plexus that release NO and VIP. Without them, the LES fails to relax and has high resting tone, and peristalsis in the distal esophagus is lost. Dysphagia to solids and liquids from the onset suggests a motility problem rather than a mechanical obstruction.

D. Anti-AChR antibodies cause myasthenia gravis (oropharyngeal dysphagia, fatigable weakness).

E. Smooth-muscle fibrosis and atrophy describes scleroderma, which lowers LES tone instead.

19. E. Clonorchis sinensis

Biliary malignancies · Step 1

A perihilar mass at the hepatic duct confluence (Klatskin tumor) is a cholangiocarcinoma. Liver flukes such as Clonorchis sinensis, acquired from raw freshwater fish, cause chronic bile duct inflammation and raise the risk. PSC and choledochal cysts are other risk factors.

D. Schistosoma haematobium is linked to squamous cell carcinoma of the bladder.

20. A. Immunity from vaccination

Infectious liver injury · Step 1

The hepatitis B vaccine contains only recombinant HBsAg, so a vaccinated person makes anti-HBs alone. Anti-HBc appears only after natural infection, because the vaccine contains no core antigen.

C. Past natural infection would also give a positive (IgG) anti-HBc.

21. C. No further test is needed

Acute pancreatitis · Step 2

Acute pancreatitis is diagnosed with 2 of 3 criteria: typical epigastric pain, lipase at least 3 times normal, and typical imaging findings. He already meets two. Early CT is not needed and can miss necrosis; it is reserved for an unclear diagnosis or for patients who fail to improve after 48–72 hours. A RUQ ultrasound is still done to look for gallstones.

A. CT within the first 48–72 hours adds little when the diagnosis is already clear and can underestimate necrosis.

22. B. 8-week trial of a proton pump inhibitor

GERD · Step 2

Typical GERD without alarm features is diagnosed clinically and treated with an empiric PPI trial (plus lifestyle changes). Endoscopy is indicated for alarm features (dysphagia, weight loss, bleeding, anemia), failure of PPI therapy, or Barrett screening in high-risk patients.

23. A. "Corkscrew" esophagus

Esophageal motility disorders · Step 1

Diffuse esophageal spasm causes disorganized contractions that give a "corkscrew" or "rosary bead" appearance. It may respond to calcium channel blockers or nitrates; dilation or botulinum toxin are other options.

E. An outpouching above the UES is a Zenker diverticulum.

24. D. Retrocecal, against the iliopsoas muscle

Appendicitis · Step 1

The psoas sign (pain on passive hip extension) means a retrocecal appendix is irritating the iliopsoas. The obturator sign (pain on internal rotation of the flexed hip) suggests a pelvic appendix. Pain migrating from the umbilicus to the RLQ reflects visceral pain becoming parietal as inflammation reaches the peritoneum.

C. A pelvic appendix gives a positive obturator sign.

25. E. Elective laparoscopic cholecystectomy

Cholelithiasis · Step 2

Recurrent biliary colic (transient cystic duct obstruction, normal LFTs, no fever) is an indication for elective laparoscopic cholecystectomy. Asymptomatic gallstones need no treatment, but symptomatic ones do, because attacks recur and complications such as cholecystitis can follow.

A. Reassurance alone is for asymptomatic gallstones.

D. ERCP removes common bile duct stones; her normal LFTs argue against choledocholithiasis.

26. E. Emergency laparotomy

Traumatic liver injury · Step 2

A hemodynamically unstable trauma patient with a positive FAST needs emergency laparotomy to control bleeding. Unstable patients should not be sent to CT.

27. C. Encephalopathy and INR of 1.5 or more within 26 weeks, with no prior liver disease

Acute liver failure and cirrhosis · Step 1

Acute liver failure is severe acute liver injury with encephalopathy and an INR of 1.5 or more, developing within 26 weeks in someone without pre-existing liver disease. Patients need ICU care and early transplant referral.

A. Encephalopathy on a background of cirrhosis is decompensated chronic liver disease, not acute liver failure.

28. C. Right lower quadrant ultrasound

Appendicitis · Step 2

In children and pregnant patients, ultrasound is the first imaging test for suspected appendicitis to avoid radiation. CT with contrast is the most accurate test in adults, or if ultrasound is inconclusive.

29. A. Eosinophilic esophagitis

Esophagitis · Step 1

Eosinophilic esophagitis occurs in atopic patients (asthma, allergic rhinitis, atopic dermatitis) and is driven by food allergens. Classic findings are food impaction, "trachealized" rings and furrows, and at least 15 eosinophils/hpf. Management includes eliminating food allergens, PPIs, and swallowed topical steroids.

C. A Schatzki ring is a single distal mucosal ring and does not explain dense eosinophilic infiltration.

E. GERD can cause some eosinophils, but rings, furrows, atopy, and food impaction point to EoE.

30. B. Cholangiocarcinoma

Autoimmune liver injury · Step 1

PSC carries a high lifetime risk of cholangiocarcinoma, often presenting as a new dominant stricture with rising CA 19-9, jaundice, and weight loss. This cancer risk is specific to PSC, not PBC. PSC also raises the risk of gallbladder and colorectal cancer.

31. A. H. pylori eradication therapy

PUD · Step 2

Chronic H. pylori infection drives gastric MALT lymphoma. Most low-grade, localized tumors regress after H. pylori eradication, so that is first-line therapy. Radiation or chemotherapy is reserved for persistent or advanced disease.

32. C. Therapeutic phlebotomy

Metabolic liver injury · Step 2

Regular phlebotomy is first-line: each unit of blood removes iron, and it is continued until ferritin is in the target range. Chelation is reserved for patients who cannot tolerate phlebotomy, such as those with anemia or heart failure.

D. D-penicillamine chelates copper and is used for Wilson disease.

33. A. JAK2 V617F mutation testing

Vascular liver injury · Step 1

Myeloproliferative neoplasms, especially polycythemia vera, are the most common underlying cause of Budd-Chiari syndrome. Most patients carry the JAK2 V617F mutation. Aquagenic pruritus and a high hemoglobin are clues.

34. A. Vitamin B12

Malabsorption · Step 1

The terminal ileum absorbs vitamin B12 (bound to intrinsic factor) and bile salts. Resection causes B12 deficiency (megaloblastic anemia) and bile salt malabsorption, which can lead to fat malabsorption and diarrhea.

C. Folate is absorbed in the jejunum.

D. Iron is absorbed in the duodenum.

35. A. Urease

PUD · Step 1

H. pylori urease converts urea to ammonia, which neutralizes acid around the bacterium. This is also the basis of the urea breath test and the rapid urease (CLO) test.

36. E. Smoking cessation and IV A1AT augmentation

Metabolic liver injury · Step 2

Smoking cessation is the single most important step, as smoke inactivates the remaining A1AT. IV augmentation therapy raises A1AT levels in patients with emphysema. Liver transplant corrects the defect but is reserved for advanced liver disease.

37. B. PPI + amoxicillin + clarithromycin

PUD · Step 2

First-line H. pylori eradication is triple therapy: PPI + amoxicillin + clarithromycin, bd for 7–14 days. Metronidazole replaces amoxicillin in penicillin allergy.

Additional information: Some guidelines (e.g. the 2024 ACG guideline in the US) prefer bismuth quadruple therapy (PPI + bismuth + metronidazole + tetracycline) as first line where clarithromycin resistance is high or after previous macrolide exposure.

38. D. Maintain perfusion, treat the infection, and trend LFTs

Vascular liver injury · Step 2

Ischemic hepatitis is managed by restoring and maintaining perfusion and treating the cause of shock. Aminotransferases peak within 1–3 days and typically fall rapidly over 7–10 days once blood flow is restored.

A. Transplant evaluation is for acute liver failure with encephalopathy and worsening coagulopathy; she is alert with a near-normal INR.

39. E. Pill-induced esophagitis

Esophagitis · Step 2

Doxycycline is a classic cause of pill esophagitis, especially when swallowed with little water before lying down. The pill sits in the esophagus and injures the mucosa directly. Treatment is to stop or change the drug, and to take tablets with a full glass of water while upright.

40. A. Esophageal manometry

Esophageal motility disorders · Step 2

Esophageal manometry is the gold standard. In achalasia it shows incomplete LES relaxation, raised LES pressure, and absent distal peristalsis. Endoscopy is done first to exclude malignancy at the GE junction (pseudoachalasia).

41. E. MRCP or endoscopic ultrasound

Choledocholithiasis · Step 2

With an intermediate likelihood of a CBD stone (abnormal LFTs or a dilated duct, but no stone seen and no cholangitis), less invasive imaging with MRCP or EUS comes first. ERCP is reserved for high likelihood or a confirmed stone, because it carries risks such as pancreatitis.

42. E. Non-occlusive mesenteric ischemia

Intestinal ischemia · Step 1

Non-occlusive mesenteric ischemia follows low cardiac output and splanchnic vasoconstriction (shock, vasopressors, digoxin), without a blockage in the vessel. Treatment focuses on improving perfusion; intra-arterial vasodilators may help.

43. C. Irreversible inhibition of the H+/K+ ATPase in parietal cells

GERD · Step 1

PPIs irreversibly inhibit the H+/K+ ATPase (proton pump), the final step of acid secretion. They are activated in acid, so they work best taken before a meal, when pumps are active.

A. Competitive H2-receptor blockade describes famotidine and other H2 blockers.

E. A protective barrier over ulcers describes sucralfate.

44. B. Supportive care: bowel rest, IV fluids, and correcting hypotension

Intestinal ischemia · Step 2

Most ischemic colitis is non-gangrenous and resolves with supportive care: bowel rest, IV fluids, improving perfusion (avoiding vasoconstrictors) and ABX in moderate to severe cases. Colonoscopy confirms the diagnosis. Surgery is for gangrene, perforation, or deterioration.

45. B. Upper endoscopy with biopsy

GERD · Step 2

Dysphagia, weight loss, and iron-deficiency anemia are alarm features that require endoscopy with biopsy to exclude malignancy, rather than an empiric PPI trial.

46. E. Splenic flexure

Intestinal ischemia · Step 1

Ischemic colitis from hypoperfusion affects watershed areas. The splenic flexure (between the SMA and IMA territories) and the rectosigmoid junction are most vulnerable. The distal rectum is spared because it has a dual blood supply from the IMA and the iliac branches.

D. The distal rectum has a dual blood supply and is usually spared.

47. D. Surreptitious stimulant laxative use

Diarrhea · Step 1

Melanosis coli (brown-black pigmentation from lipofuscin in macrophages) is caused by long-term anthraquinone laxatives such as senna. Stimulant laxatives cause secretory diarrhea, so the osmotic gap is low.

B. Polyethylene glycol is an osmotic laxative, causing a high osmotic gap and no pigmentation.

48. C. Pseudohyphae and budding yeast

Esophagitis · Step 1

Candida esophagitis, the most common infectious esophagitis, shows pseudohyphae and budding yeast. Inhaled steroids, ABX, diabetes, and immunosuppression are risk factors.

A. Owl's eye inclusions suggest CMV.

D. Goblet cells in columnar epithelium describe Barrett esophagus.

E. Multinucleated giant cells suggest HSV.

49. A. Prostaglandin E2

Gastritis · Step 1

NSAIDs inhibit COX-1, lowering prostaglandins (PGE2) that normally increase mucus and bicarbonate secretion, maintain mucosal blood flow, and reduce acid secretion. Losing that protection allows acid injury.

50. C. Inhibition of xanthine oxidase, which normally breaks down 6-mercaptopurine

IBD · Step 1

Azathioprine is converted to 6-mercaptopurine, which is broken down by xanthine oxidase (and TPMT). Allopurinol blocks xanthine oxidase, so 6-MP accumulates and causes bone marrow suppression. The azathioprine dose must be greatly reduced if both are used.

51. D. Decreased hepcidin, leading to increased ferroportin activity

Metabolic liver injury · Step 1

HFE mutations reduce hepcidin production. Hepcidin normally binds ferroportin and causes it to be degraded; without it, ferroportin keeps exporting iron from duodenal cells and macrophages, so iron absorption rises unchecked. Iron deposits in the liver, pancreas, heart, skin, and joints.

B. High hepcidin blocks iron export; this is the mechanism of anemia of chronic disease.

E. Transfusions cause secondary iron overload, but this patient has a genetic cause.

52. D. Decreased gastric acid; increased serum gastrin

Gastritis · Step 1

Parietal cell loss causes achlorhydria. Without acid there is no negative feedback on antral G cells, so gastrin rises (G-cell hyperplasia). Chronic hypergastrinemia also raises the risk of gastric carcinoid tumors.

53. E. Colonoscopy 6–8 weeks after recovery

Diverticular diseases · Step 2

Colonoscopy is deferred until the inflammation settles (about 6–8 weeks) and is then performed to exclude colorectal cancer, which can mimic diverticulitis on CT.

54. D. Ursodeoxycholic acid

Autoimmune liver injury · Step 2

Ursodeoxycholic acid is a hydrophilic bile acid that reduces bile salt toxicity, improves LFTs, and slows progression in PBC. Cholestyramine is used for pruritus. Transplant is for end-stage disease.

E. Cholestyramine relieves pruritus by binding bile salts in the gut; it does not slow disease progression.

55. E. Ulcerative colitis

Autoimmune liver injury · Step 1

Most patients with PSC have inflammatory bowel disease, usually ulcerative colitis, which may be mild or silent. A colonoscopy is recommended at diagnosis, and surveillance follows because the combination raises colorectal cancer risk.

56. E. Activated charcoal, and measure a paracetamol level 4 hours after ingestion

Iatrogenic and toxic liver injury · Step 2

She has taken about 333 mg/kg, well above the toxic dose. Activated charcoal helps within 1–2 hours of ingestion. A 4-hour paracetamol level plotted on the Rumack-Matthew nomogram guides NAC; given the dose, NAC is often started without waiting. Early LFTs can be normal despite serious toxicity.

57. E. Giardia lamblia

Diarrhea · Step 1

Giardia (from contaminated river or stream water) causes non-inflammatory, fatty, foul-smelling diarrhea and bloating. Trophozoites attach to the duodenal mucosa, causing malabsorption without invasion. It is treated with metronidazole or tinidazole.

C. E. histolytica invades the mucosa, causing bloody diarrhea (dysentery) and liver abscesses.

58. A. PPI + clarithromycin + metronidazole

PUD · Step 2

In penicillin allergy, metronidazole replaces amoxicillin in triple therapy, giving PPI + clarithromycin + metronidazole.

E. Amoxicillin is a penicillin and is contraindicated after anaphylaxis.

59. C. Cigarette smoking

IBD · Step 1

Smoking increases the risk and severity of Crohn disease but is associated with a lower risk of ulcerative colitis. Family history and a Western diet raise the risk of both. NOD2 mutations are linked to Crohn disease, and HLA-DRB1 to UC.

60. A. Interferon-gamma release assay (or tuberculin skin test)

IBD · Step 2

Infliximab is a TNF-α inhibitor. TNF-α is essential for maintaining granulomas, so blocking it can reactivate latent tuberculosis. Screen with an IGRA or tuberculin skin test (and for hepatitis B) before starting.

C. TPMT activity is checked before azathioprine or 6-MP, not before anti-TNF therapy.

61. B. Rotavirus

Diarrhea · Step 1

Rotavirus is the classic cause of severe watery diarrhea in unvaccinated infants and young children, especially in childcare settings. It damages villi, causing malabsorption. A live oral vaccine prevents it.

62. E. Lifestyle changes aiming for 7–10% weight loss

Metabolic liver injury · Step 2

Weight loss through diet and exercise is the mainstay of MASLD treatment: 7–10% weight loss can reduce steatosis, inflammation, and even fibrosis. Metabolic risk factors should also be treated. Drug therapy (e.g. GLP-1 receptor agonists or resmetirom) is considered for MASH with significant fibrosis.

63. D. Vibrio vulnificus

Metabolic liver injury · Step 1

Iron overload favors iron-loving (siderophilic) bacteria such as Vibrio vulnificus, Yersinia, and Listeria. Vibrio vulnificus from raw shellfish can cause fulminant sepsis with hemorrhagic bullae, so patients are told to avoid raw shellfish.

64. B. Rifaximin

Acute liver failure and cirrhosis · Step 2

Rifaximin, a poorly absorbed ABX, reduces ammonia-producing gut bacteria and is added to lactulose to prevent recurrent hepatic encephalopathy.

E. Benzodiazepines can precipitate or worsen encephalopathy.

65. D. Echocardiography

Vascular liver injury · Step 2

Hepatomegaly with jugular venous distention and a positive hepatojugular reflux points to congestive hepatopathy from right heart failure, so echocardiography is the key test. Budd-Chiari syndrome causes hepatomegaly and ascites without JVD.

66. A. HLA-DQ2 and DQ8

Malabsorption · Step 1

Celiac disease is associated with HLA-DQ2 and DQ8. Gliadin (from wheat, rye, and barley) is deamidated by tissue transglutaminase and presented on these HLA molecules, driving a T-cell response that damages the mucosa. The rash is dermatitis herpetiformis. Iron deficiency reflects duodenal involvement.

C. HLA-B27 is linked to the seronegative spondyloarthropathies.

D. HLA-DR3/DR4 is classically linked to type 1 diabetes.

67. B. Oral rehydration, with loperamide if needed

Diarrhea · Step 2

Mild traveler's diarrhea, most often from enterotoxigenic E. coli, is self-limited. Oral rehydration is the mainstay, with loperamide for symptom control if there is no fever or blood. ABX are reserved for moderate to severe cases.

68. D. Pregnant patients

PUD · Step 1

Misoprostol (a PGE1 analog) increases mucus and bicarbonate and reduces acid. It also causes uterine contractions and can induce abortion, so it is contraindicated in pregnancy. Its other main side effect is diarrhea.

69. A. Bile salts entering the colon directly

Biliary fistulas · Step 1

A cholecystocolic fistula lets bile bypass the small intestine, where bile salts are normally reabsorbed. Bile salts reaching the colon stimulate water secretion, causing bile acid diarrhea; bile also fails to reach the small bowel, so fat absorption suffers.

70. E. Dry mouth and constipation

IBS · Step 1

TCAs block muscarinic receptors, causing dry mouth, constipation, urinary retention, and blurred vision. They also cause sedation (via H1) and orthostasis (via α1). The constipating effect can be useful in IBS-D, and TCAs reduce visceral hypersensitivity.

71. E. IgG-based celiac serology

Malabsorption · Step 2

Selective IgA deficiency is more common in celiac disease and makes IgA-based tests falsely negative. IgG-based serology (deamidated gliadin peptide or tTG) is used instead, followed by duodenal biopsy if positive.

72. A. IV octreotide and IV ABX

Acute liver failure and cirrhosis · Step 2

For suspected variceal bleeding, start octreotide (a splanchnic vasoconstrictor that lowers portal pressure) and prophylactic ABX (e.g. ceftriaxone, which reduce infection and mortality), then perform urgent endoscopic band ligation. Beta-blockers are for prevention, not an acute bleed.

B. Non-selective beta-blockers prevent variceal bleeding but are not used in acute hemorrhage with hypotension.

73. C. Intramuscular vitamin B12

Gastritis · Step 2

Autoimmune gastritis destroys parietal cells, so intrinsic factor is lost and B12 cannot be absorbed normally. Parenteral (IM) B12 replacement bypasses the problem; high-dose oral B12 is an alternative.

A. Folic acid alone can correct the anemia while the neurologic damage from B12 deficiency progresses.

74. A. IV third-generation cephalosporin plus IV albumin

Acute liver failure and cirrhosis · Step 2

An ascitic neutrophil count of 250/μL or more diagnoses spontaneous bacterial peritonitis. Treat with a third-generation cephalosporin (e.g. ceftriaxone or cefotaxime). IV albumin reduces the risk of hepatorenal syndrome. Long-term ABX prophylaxis follows.

D. SBP has no surgical source; laparotomy is for secondary peritonitis such as a perforation.

75. C. Increased stool osmotic gap

Malabsorption · Step 1

Undigested lactose draws water into the bowel, causing osmotic diarrhea with a high stool osmotic gap (over 125 mOsm/kg) that stops with fasting. Colonic bacteria ferment the lactose to hydrogen gas and acids, which explains the bloating and the positive breath test.

A. A low osmotic gap (under 50) points to secretory diarrhea.

76. A. Bacillus cereus

Diarrhea · Step 1

B. cereus spores survive cooking. In rice left at room temperature they germinate and form a preformed, heat-stable emetic toxin (cereulide), causing vomiting within 1–6 hours. This is called "reheated rice syndrome".

B. S. aureus also causes rapid vomiting from a preformed toxin, but the classic sources are mayonnaise, potato salad, and custards.

77. D. IV ABX and laparoscopic cholecystectomy during this admission

Cholecystitis · Step 2

This is acute calculous cholecystitis: prolonged RUQ pain, fever, Murphy sign, leukocytosis, typical ultrasound findings, and normal LFTs. Treat with supportive care and ABX, followed by early laparoscopic cholecystectomy, ideally within 72 hours.

B. ERCP is for common bile duct stones; normal bilirubin and ALP make this unlikely.

E. Cholecystostomy is reserved for patients too unwell for surgery or with acalculous cholecystitis.

78. C. Hepatopancreatic ampulla (of Vater)

Choledocholithiasis · Step 1

Gallstone pancreatitis occurs when a stone blocks the ampulla, where the common bile duct and pancreatic duct meet, causing pancreatic enzyme activation. A cholestatic LFT pattern with high lipase and gallstones on ultrasound suggests a gallstone cause; it is the most common cause of acute pancreatitis.

B. A cystic duct stone causes biliary colic or cholecystitis with normal LFTs and lipase.

79. D. Increased NADH:NAD+ ratio

Iatrogenic and toxic liver injury · Step 1

Alcohol breakdown generates excess NADH. A high NADH:NAD+ ratio inhibits fatty acid beta-oxidation and favors fatty acid synthesis and glycerol-3-phosphate production, so triglycerides build up in hepatocytes. It also inhibits gluconeogenesis, which is why heavy drinkers can become hypoglycemic.

Test C answers and explanations

1 **C**	2 **B**	3 **C**	4 **A**	5 **E**	6 **D**	7 **C**	8 **C**	9 **C**	10 **A**
11 **E**	12 **C**	13 **A**	14 **A**	15 **D**	16 **A**	17 **A**	18 **D**	19 **A**	20 **E**
21 **C**	22 **A**	23 **B**	24 **B**	25 **C**	26 **C**	27 **E**	28 **B**	29 **D**	30 **E**
31 **C**	32 **D**	33 **A**	34 **B**	35 **D**	36 **D**	37 **E**	38 **B**	39 **B**	40 **C**
41 **B**	42 **E**	43 **D**	44 **A**	45 **E**	46 **A**	47 **C**	48 **B**	49 **E**	50 **A**
51 **B**	52 **D**	53 **B**	54 **A**	55 **C**	56 **B**	57 **D**	58 **A**	59 **A**	60 **B**
61 **A**	62 **E**	63 **B**	64 **D**	65 **E**	66 **D**	67 **D**	68 **E**	69 **E**	70 **B**
71 **E**	72 **C**	73 **C**	74 **B**	75 **C**	76 **D**	77 **D**	78 **E**	79 **D**	

1. C. No further test is needed

Liver tumors · Step 2

In a patient with cirrhosis, a lesion with arterial-phase enhancement and washout on multiphase CT or MRI is diagnostic of hepatocellular carcinoma; biopsy is not required. Biopsy is reserved for lesions with atypical imaging.

2. B. Blocking acetylcholine release from nerve terminals

Esophageal motility disorders · Step 1

Botulinum toxin cleaves SNARE proteins and blocks acetylcholine release from excitatory neurons, lowering LES tone. The effect wears off over months, so it is used in patients who are poor candidates for dilation or myotomy.

C. Blocking L-type calcium channels is how calcium channel blockers such as nifedipine work.

3. C. IV ABX and percutaneous drainage of the abscess

Appendicitis · Step 2

A contained appendiceal abscess in a stable patient is treated with IV ABX and percutaneous drainage; operating on the inflamed tissue early carries a higher risk of complications. Appendectomy may follow later. Generalized peritonitis, by contrast, needs emergency surgery.

4. A. Transferrin saturation and serum ferritin

Metabolic liver injury · Step 2

Transferrin saturation (above 45% is abnormal) with ferritin is the initial screen for iron overload. HFE genetic testing follows, which is also offered directly to first-degree relatives. Liver MRI or biopsy is reserved for unclear cases.

5. E. Anion-gap metabolic acidosis from lactate

Intestinal ischemia · Step 1

Ischemic bowel switches to anaerobic metabolism, producing lactic acid and an anion-gap metabolic acidosis. A rising lactate is a late and worrying sign that suggests infarction.

6. D. Tropheryma whipplei

Malabsorption · Step 1

Whipple disease is caused by Tropheryma whipplei and classically affects middle-aged men. Features include arthralgias, cardiac and neurologic symptoms, diarrhea, and weight loss, with PAS-positive foamy macrophages on biopsy. It is treated with long-term ABX.

E. MAC can also produce foamy macrophages in AIDS, but it is acid-fast positive and the patient here is not immunocompromised.

7. C. Trypanosoma cruzi

Esophageal motility disorders · Step 1

Chagas disease (T. cruzi, spread by the reduviid "kissing" bug in Latin America) destroys myenteric neurons. This causes secondary achalasia, megacolon, and dilated cardiomyopathy.

A. T. brucei causes African sleeping sickness.

8. C. Sigmoid colon, because it has the highest intraluminal pressure

Diverticular diseases · Step 1

The sigmoid colon has the narrowest lumen and generates the highest pressures (law of Laplace), especially with low-fiber diets and constipation, so diverticula form there most often in Western populations.

9. C. IV ABX and percutaneous cholecystostomy

Cholecystitis · Step 2

Acalculous cholecystitis occurs in critically ill patients from gallbladder stasis and ischemia, and carries a high risk of gangrene and perforation. Unstable patients are treated with ABX and percutaneous drainage (cholecystostomy).

10. A. Copper chelation with trientine or D-penicillamine

Metabolic liver injury · Step 2

Initial treatment is copper chelation (trientine or D-penicillamine) to remove stored copper, together with a low-copper diet. Zinc, which blocks intestinal copper absorption, is used for maintenance. Transplant is reserved for acute liver failure or end-stage disease.

D. Phlebotomy removes iron, not copper; it treats hemochromatosis.

11. E. Elimination of food allergens from the diet

Esophagitis · Step 2

Eosinophilic esophagitis is an allergic, food antigen-driven inflammation of the esophagus. Elimination diets remove the trigger; PPIs and swallowed topical corticosteroids are also first-line options.

A. Pneumatic dilation of the LES treats achalasia.

12. C. Very high LDH, with aminotransferases that fall rapidly over days

Vascular liver injury · Step 1

Ischemic hepatitis causes a sharp rise in aminotransferases within 1–3 days of hypotension, with a very high LDH, followed by a rapid fall once perfusion is restored. Viral hepatitis rises and falls more slowly.

A. An AST:ALT ratio above 2 with AST under 300 suggests alcoholic liver disease.

13. A. Duodenal ulcer

PUD · Step 1

Duodenal ulcer pain improves with meals (food buffers acid and stimulates bicarbonate) and is worse when fasting or at night, so patients may gain weight. Gastric ulcer pain worsens with eating, so patients may lose weight.

B. Gastric ulcer pain worsens with meals and is associated with weight loss.

14. A. Prophylactic PPI or H2 blocker

Gastritis · Step 2

Critically ill patients, especially those on mechanical ventilation, with coagulopathy, burns, or head injury (Cushing ulcers), are at risk of stress ulcer bleeding. Acid suppression with a PPI or H2 blocker is used as prophylaxis.

15. D. NS5B RNA-dependent RNA polymerase

Infectious liver injury · Step 1

Sofosbuvir is a nucleotide analog that inhibits the NS5B RNA-dependent RNA polymerase. Direct-acting antiviral combinations (e.g. sofosbuvir with an NS5A inhibitor) cure over 95% of HCV infections.

A. Reverse transcriptase inhibitors such as tenofovir are used for HBV and HIV.

E. NS3/4A protease inhibitors end in -previr, e.g. glecaprevir.

16. A. Metronidazole, then luminal paromomycin

Diarrhea · Step 1

Entamoeba histolytica causes amebic dysentery (flask-shaped colonic ulcers) and amebic liver abscess. Trophozoites with ingested RBCs are characteristic. Treat with metronidazole, followed by a luminal agent to clear cysts.

17. A. Prolonged PT; normal platelet count

Malabsorption · Step 1

Fat malabsorption reduces absorption of fat-soluble vitamins (A, D, E, K). Vitamin K deficiency impairs gamma-carboxylation of factors II, VII, IX, and X. Factor VII has the shortest half-life, so PT prolongs first. Platelets are unaffected.

18. D. Cholestyramine

Autoimmune liver injury · Step 1

Cholestyramine is a bile acid sequestrant. It interrupts enterohepatic circulation, lowering bile salts that build up in cholestasis and cause pruritus. Give it apart from other drugs (including UDCA), because it binds them too.

C. Ezetimibe blocks cholesterol absorption (NPC1L1) and is not used for pruritus.

19. A. Increased unconjugated bilirubin from chronic hemolysis

Cholelithiasis · Step 1

Chronic hemolysis (sickle cell disease, hereditary spherocytosis) increases bilirubin production. The extra bilirubin precipitates with calcium in the gallbladder as black pigment stones, which are often radiopaque.

C. Bacterial deconjugation produces brown pigment stones, which form in the bile ducts.

20. E. Endovascular angioplasty and stenting

Intestinal ischemia · Step 2

Symptomatic chronic mesenteric ischemia is treated with revascularization, usually endovascular angioplasty and stenting (or open bypass), plus risk factor control. It relieves pain and prevents progression to acute bowel infarction.

21. C. Phrenic nerve

Traumatic liver injury · Step 1

Blood or bile under the right hemidiaphragm irritates the diaphragmatic peritoneum, which is supplied by the phrenic nerve (C3–C5). Pain is referred to the C4 dermatome over the shoulder.

22. A. Hepatitis C

Infectious liver injury · Step 1

Chronic HCV is the main cause of mixed cryoglobulinemia: immune complexes deposit in small vessels and glomeruli, causing vasculitis (palpable purpura), arthralgia and MPGN. HBV is classically linked to polyarteritis nodosa and membranous nephropathy.

23. B. Stop omeprazole 2 weeks before a urea breath test, done at least 4 weeks after ABX

PUD · Step 2

PPIs, ABX, and bismuth suppress H. pylori and cause false-negative breath, stool antigen, and rapid urease tests. Stop PPIs 2 weeks before testing, and wait at least 4 weeks after ABX and bismuth.

C. Bismuth suppresses H. pylori and would cause a false-negative result.

D. Serology stays positive after eradication, so it cannot confirm cure.

24. B. Hemobilia

Traumatic liver injury · Step 2

Hemobilia is bleeding into the biliary tree, often from a post-traumatic arterial pseudoaneurysm. The classic triad is RUQ pain, jaundice, and GI bleeding. It is usually treated by angioembolization.

25. C. Hypovolemia leading to gastric mucosal ischemia

Gastritis · Step 1

Curling ulcers follow severe burns. Massive fluid loss causes hypovolemia and gastric mucosal ischemia, which leads to erosions. Burn = Curling ("cooking"); brain = Cushing.

D. Increased vagal stimulation explains Cushing ulcers after brain injury or raised ICP.

26. C. Repeat endoscopy every 3–5 years

GERD · Step 2

Barrett esophagus without dysplasia is followed with endoscopic surveillance every 3–5 years to detect dysplasia. Dysplasia shortens the interval and may need endoscopic eradication therapy.

27. E. Shigella

Diarrhea · Step 1

Shigella has a very low infectious dose (as few as 10 organisms), so it spreads easily person to person. It invades colonic mucosa, causing dysentery, and is non-motile and does not produce H_2S.

D. Salmonella is motile, produces H_2S, and usually comes from poultry or eggs.

28. B. Fibromyalgia

IBS · Step 1

IBS is a disorder of gut-brain interaction driven by visceral hypersensitivity, altered microbiome, and abnormal motility. It commonly coexists with fibromyalgia, anxiety, and depression.

29. D. Portal hypertension

Acute liver failure and cirrhosis · Step 1

The serum-ascites albumin gradient (SAAG) is 3.0 − 1.1 = 1.9 g/dL. A SAAG of 1.1 or more indicates portal hypertension (cirrhosis, heart failure, Budd-Chiari). Below 1.1 suggests peritoneal causes such as cancer, TB, or pancreatitis. Neutrophils under 250/μL argue against SBP.

30. E. Microvesicular steatosis

AFLP · Step 1

AFLP shows microvesicular steatosis: tiny fat droplets fill hepatocytes without displacing the nucleus, reflecting mitochondrial failure of fatty acid oxidation.

A. Macrovesicular steatosis with Mallory-Denk bodies suggests alcoholic liver disease.

D. Periportal fibrin deposits and hemorrhage suggest HELLP or pre-eclampsia.

31. C. IgA anti-tTG antibodies with total serum IgA

Malabsorption · Step 2

Screen for celiac disease with IgA anti-tTG while the patient is still eating gluten. Check total IgA because IgA deficiency (more common in celiac disease) gives false-negative results. Confirm with EGD and duodenal biopsy.

B. Biopsy confirms the diagnosis but follows a positive serology.

32. D. Bacterial deconjugation of bilirubin within the bile duct

Choledocholithiasis · Step 1

Brown pigment stones form within the bile ducts (primary CBD stones), where infection with bacteria producing β-glucuronidase deconjugates bilirubin, which precipitates with calcium. They can form years after cholecystectomy.

33. A. Biopsy the ulcer margins

PUD · Step 2

Gastric ulcers can be malignant, so they are biopsied (along with testing for H. pylori), and healing is often confirmed with repeat endoscopy. Duodenal ulcers are rarely malignant and are not routinely biopsied.

34. B. Sliding hiatal hernia

GERD · Step 1

In a sliding hiatal hernia, the most common type, the GE junction itself moves above the diaphragm, weakening the antireflux barrier and causing GERD.

D. In a paraesophageal hernia, the GE junction stays in place and part of the stomach rolls up beside it; it risks strangulation rather than reflux.

35. D. Splanchnic vasodilation causing renal vasoconstriction

Acute liver failure and cirrhosis · Step 1

Hepatorenal syndrome is functional kidney failure. Portal hypertension causes splanchnic vasodilation (nitric oxide), lowering effective arterial volume. RAAS and sympathetic activation then constrict the renal arteries. The kidneys are structurally normal and avidly retain sodium.

36. D. Gallbladder carcinoma

Biliary malignancies · Step 1

A porcelain gallbladder (calcified wall from chronic inflammation) is associated with gallbladder carcinoma, as are large gallstones, polyps over 1 cm, and PSC. Cholecystectomy is often recommended.

37. E. CT angiography of the abdomen

Intestinal ischemia · Step 2

Pain out of proportion to examination in a patient with an embolic source suggests acute mesenteric ischemia. CT angiography is the diagnostic test of choice. Treatment is urgent revascularization (endovascular or surgical) with anticoagulation, plus resection of necrotic bowel.

38. B. Loperamide

IBS · Step 2

Loperamide, a peripherally acting μ-opioid agonist, slows gut transit and treats diarrhea in IBS-D. Laxatives are used for constipation-predominant IBS.

39. B. Hepatitis E virus

Infectious liver injury · Step 1

HEV spreads by the fecal-oral route (contaminated water) and is usually self-limited, but in pregnancy, particularly the third trimester, it can cause fulminant hepatitis with high mortality. It is a naked +ssRNA hepevirus.

D. HAV is also fecal-oral, but it is not associated with fulminant hepatitis in pregnancy.

40. C. Doppler ultrasound of the hepatic veins

Vascular liver injury · Step 2

RUQ pain, hepatomegaly, and ascites in a patient with a myeloproliferative neoplasm suggest Budd-Chiari syndrome (hepatic vein thrombosis). Doppler ultrasound of the hepatic veins is the first-line test; CT or MR venography confirms it. A normal JVP argues against a cardiac cause.

D. Echocardiography evaluates congestive hepatopathy, but her normal JVP and lack of edema argue against heart failure.

E. A HIDA scan evaluates acute cholecystitis.

41. B. Vipoma

Diarrhea · Step 1

A low osmotic gap (under 50) and diarrhea that persists with fasting indicate secretory diarrhea. A vipoma causes WDHA syndrome: watery diarrhea, hypokalemia, and achlorhydria. Other hormonal causes are gastrinoma and carcinoid syndrome.

A. Lactose intolerance is osmotic: high gap, and it stops with fasting.

C. PEG is an osmotic laxative and produces a high osmotic gap.

42. E. Tetracycline and folic acid

Malabsorption · Step 2

Tropical sprue follows prolonged stay in the tropics and causes mucosal injury with malabsorption, often with folate (and B12) deficiency. It is treated with ABX (tetracycline) plus folic acid.

43. D. Gastroduodenal artery

PUD · Step 1

Posterior duodenal ulcers can erode into the gastroduodenal artery, which runs behind the first part of the duodenum, causing major hemorrhage. Anterior duodenal ulcers are more likely to perforate.

E. The left gastric artery is typically involved by ulcers on the lesser curvature of the stomach.

44. A. Cholecystectomy

Ascending cholangitis · Step 2

The gallbladder is the source of most CBD stones, so cholecystectomy (once she has recovered, usually during the same admission) prevents recurrent choledocholithiasis and cholangitis.

45. E. Superior mesenteric artery

Intestinal ischemia · Step 1

The SMA leaves the aorta at an acute angle and carries high flow, so emboli lodge there most often. This causes ischemia of the small bowel and proximal colon.

46. A. Adenocarcinoma of the distal esophagus

GERD · Step 1

Barrett esophagus is intestinal metaplasia, in which nonciliated columnar epithelium with goblet cells replaces stratified squamous epithelium. It can progress through dysplasia to adenocarcinoma of the distal esophagus, so patients need endoscopic surveillance.

B. Squamous cell carcinoma is linked to smoking and alcohol and typically arises in the upper or middle esophagus.

47. C. Mesalazine (5-ASA)

IBD · Step 2

5-ASA drugs (mesalazine, sulfasalazine) are first-line for inducing and maintaining remission in mild-to-moderate ulcerative colitis. Corticosteroids are for flares and are not used for maintenance.

B. Long-term steroids cause serious adverse effects and do not maintain remission.

48. B. Cholangiocarcinoma

IBD · Step 1

Primary sclerosing cholangitis is strongly associated with ulcerative colitis (p-ANCA positive) and greatly raises the risk of cholangiocarcinoma. It also raises colorectal cancer risk further, so patients with both need closer colonoscopic surveillance.

49. E. Centrilobular hepatocytes (zone 3)

Vascular liver injury · Step 1

Ischemic hepatitis (shock liver) causes centrilobular (zone 3) necrosis. Zone 3 hepatocytes surround the central vein and receive the least oxygenated blood, so they are the first to die when perfusion falls. A massive aminotransferase rise with a very high LDH is typical.

A. Zone 1 is closest to the portal triad and best oxygenated; it is the last affected by hypoperfusion.

C. Zone 2 necrosis is classically associated with yellow fever.

50. A. Prednisolone

Iatrogenic and toxic liver injury · Step 2

Severe alcoholic hepatitis (Maddrey discriminant function of 32 or more) is treated with prednisolone if there is no infection or active bleeding. Abstinence, nutrition, and thiamine are essential for all patients.

51. B. Hepatic vein angioplasty with or without stenting

Vascular liver injury · Step 2

Budd-Chiari treatment steps up when anticoagulation alone fails: angioplasty or stenting for short venous stenoses, then TIPS, then liver transplant for fulminant failure or cirrhosis.

52. D. IV ABX, resuscitation, and emergency laparotomy

Diverticular diseases · Step 2

Perforated diverticulitis with generalized (fecal) peritonitis requires resuscitation, broad-spectrum IV ABX, and emergency laparotomy, often a sigmoid resection with an end colostomy (Hartmann procedure).

53. B. Ingestion of preformed heat-stable enterotoxin

Diarrhea · Step 1

S. aureus multiplies in food left at room temperature and produces a heat-stable enterotoxin before the food is eaten. Because the toxin is already present, symptoms start within hours, and reheating the food does not prevent illness.

C. Inhibition of the 60S ribosomal subunit describes Shiga toxin.

54. A. Fever, right upper quadrant pain, jaundice

Ascending cholangitis · Step 1

Charcot triad is fever, right upper quadrant pain, and jaundice. Adding hypotension and altered mental status gives Reynolds pentad, which signals septic shock.

B. Painless jaundice with a palpable gallbladder suggests malignancy (Courvoisier law).

C. RUQ pain, Murphy sign, and fever without jaundice suggest acute cholecystitis.

55. C. Serum triglycerides of 2400 mg/dL

Acute pancreatitis · Step 1

Hypertriglyceridemia (usually over 1000 mg/dL) is a cause of acute pancreatitis (the H in I GET SMASHED). Lipase breaks down the triglycerides into toxic free fatty acids that injure the pancreas. Milky (lipemic) serum is a clue. Treatment includes insulin infusion to lower triglycerides.

D. Hypercalcemia (not low calcium) causes pancreatitis; low calcium is a complication of it.

56. B. Antrum

Gastritis · Step 1

H. pylori gastritis typically starts in the antrum and can spread to the body. Autoimmune gastritis, by contrast, targets the body and fundus, where parietal cells live.

57. D. CT-guided percutaneous drainage plus ABX

Diverticular diseases · Step 2

Diverticular abscesses larger than about 3–4 cm are drained percutaneously alongside ABX. Surgery is reserved for failed drainage, perforation with peritonitis, or recurrent complicated disease.

58. A. IgM anti-HAV

Infectious liver injury · Step 2

Hepatitis A spreads by the fecal-oral route, has an incubation of 2–6 weeks, and causes acute hepatitis only. IgM anti-HAV indicates acute infection; IgG indicates past infection or vaccination.

B. IgG anti-HAV is positive after past infection or vaccination, so it cannot confirm acute infection.

59. A. Laparoscopic cholecystectomy before discharge

Acute pancreatitis · Step 2

After mild gallstone pancreatitis, cholecystectomy during the same admission greatly reduces the risk of recurrent pancreatitis and other gallstone complications. ERCP is reserved for cholangitis or ongoing bile duct obstruction.

60. B. Immune reaction to drug-modified liver proteins

Iatrogenic and toxic liver injury · Step 1

Halothane hepatitis is an immune-mediated reaction: trifluoroacetylated liver proteins act as neoantigens, so severe injury usually follows repeat exposure. Fever and eosinophilia are clues to an immune drug reaction.

61. A. Ganciclovir

Esophagitis · Step 1

Large, shallow, linear ulcers with "owl's eye" inclusions in endothelial and stromal cells at the ulcer base are typical of CMV esophagitis, treated with ganciclovir (or valganciclovir). HSV causes small "volcano" ulcers with multinucleated giant cells at the ulcer edge and is treated with acyclovir.

C. Acyclovir is for HSV; CMV lacks the viral thymidine kinase that activates acyclovir.

62. E. Hepatocyte copper excretion into bile

Metabolic liver injury · Step 1

Wilson disease is caused by ATP7B mutations (autosomal recessive). ATP7B moves copper into bile and onto ceruloplasmin. Without it, copper builds up in the liver and then deposits in the basal ganglia (movement and psychiatric symptoms) and cornea (Kayser-Fleischer rings).

63. B. Bouveret syndrome

Biliary fistulas · Step 2

Bouveret syndrome is gastric outlet obstruction caused by a gallstone that has passed through a cholecystoduodenal (or cholecystogastric) fistula and lodged in the pylorus or duodenum. Endoscopic removal is tried first; surgery if it fails.

E. Mirizzi syndrome is extrinsic compression of the common hepatic duct by a stone in the cystic duct, causing jaundice.

64. D. ABX therapy

Diarrhea · Step 2

This is hemolytic uremic syndrome from Shiga toxin-producing E. coli (O157:H7). ABX, and antimotility agents such as loperamide, may increase toxin release and the risk of HUS, so they are avoided. Care is supportive: fluids, transfusion, and dialysis as needed.

65. E. Restrictive cardiomyopathy from iron deposition

Metabolic liver injury · Step 1

Iron deposition in the myocardium in hemochromatosis causes restrictive cardiomyopathy (impaired filling), which can progress to dilated cardiomyopathy and arrhythmias. Phlebotomy can improve cardiac function.

66. D. Calculate a FIB-4 score, followed by elastography if indicated

Metabolic liver injury · Step 2

Fibrosis drives outcomes in MASLD. Non-invasive scores such as FIB-4 are used first, with elastography for indeterminate or high results. Biopsy is reserved for uncertain cases.

67. D. 12-lead ECG

GERD · Step 2

Chest pain must be assumed cardiac until proven otherwise, particularly in a patient with cardiovascular risk factors. Obtain an ECG (and troponins as indicated) before attributing the pain to GERD.

68. E. Corticosteroids, with or without azathioprine

Autoimmune liver injury · Step 2

Autoimmune hepatitis type 1 typically affects young women, with a hepatocellular pattern, positive ANA and anti-smooth muscle antibody and raised IgG. It responds to immunosuppression with corticosteroids and azathioprine.

69. E. Lactulose

Acute liver failure and cirrhosis · Step 2

This is hepatic encephalopathy, often triggered by constipation, infection, bleeding, or sedatives. Lactulose acidifies the colon, trapping ammonia as ammonium for excretion, and also relieves constipation. Rifaximin is added for recurrent episodes.

B. Benzodiazepines worsen hepatic encephalopathy.

D. Strict protein restriction is no longer recommended; it worsens malnutrition.

70. B. Folate deficiency

IBD · Step 1

Sulfasalazine (sulfapyridine + 5-ASA) can cause reversible oligospermia, sulfa-type reactions (rash, agranulocytosis, hemolysis in G6PD deficiency) and folate deficiency, because it inhibits folate absorption.

71. E. Hypochloremic, hypokalemic metabolic alkalosis

PUD · Step 1

Scarring from chronic ulcers can cause gastric outlet obstruction (succussion splash). Vomiting gastric acid causes loss of HCl, giving a hypochloremic metabolic alkalosis; volume loss activates aldosterone, which worsens hypokalemia.

72. C. To confirm that the ALP is from the liver rather than bone

Autoimmune liver injury · Step 1

Alkaline phosphatase comes from liver (bile ducts), bone and placenta. A raised GGT alongside it confirms a hepatobiliary (cholestatic) source. Synthetic function is assessed with INR and albumin.

73. C. Rovsing sign

Appendicitis · Step 1

Rovsing sign is RLQ pain on palpation of the LLQ, from displacing bowel contents and stretching the inflamed peritoneum. It supports the diagnosis of appendicitis.

B. Cullen sign is periumbilical bruising from retroperitoneal bleeding.

E. Murphy sign is inspiratory arrest on RUQ palpation in cholecystitis.

74. B. Decreased ileal bile salt reabsorption, supersaturating bile with cholesterol

IBD · Step 1

Ileal disease or resection disrupts enterohepatic circulation of bile salts. The smaller bile salt pool leaves bile supersaturated with cholesterol, so cholesterol gallstones form.

75. C. Empiric oral fluconazole

Esophagitis · Step 2

Candida is the most common cause of infectious esophagitis. In an immunocompromised patient with odynophagia and oral thrush, empiric systemic fluconazole is appropriate. Endoscopy is reserved for patients who do not improve within about 72 hours.

A. Acyclovir treats HSV esophagitis (small "punched-out" ulcers); oral thrush points to Candida.

B. Ganciclovir treats CMV esophagitis (large, shallow linear ulcers).

E. EGD is indicated only if empiric antifungal therapy fails.

76. D. Pneumatic dilation of the LES

Esophageal motility disorders · Step 2

Definitive options for achalasia disrupt the LES: pneumatic dilation, or surgical myotomy (Heller or POEM). Calcium channel blockers, nitrates, and botulinum toxin are less durable and suit patients unfit for these procedures.

A. Nissen fundoplication tightens the LES and is used for GERD.

77. D. It has no epithelial lining

Acute pancreatitis · Step 1

A pancreatic pseudocyst is a collection of fluid rich in pancreatic enzymes enclosed by fibrous and granulation tissue, without an epithelial lining, forming 4 or more weeks after pancreatitis. It can cause compression, infection, rupture, or bleeding; symptomatic pseudocysts are drained.

78. E. Non-selective beta-blocker such as propranolol

Acute liver failure and cirrhosis · Step 2

Primary prophylaxis for large varices is a non-selective beta-blocker or endoscopic band ligation. Non-selective beta-blockers lower portal pressure by reducing cardiac output (β1) and causing splanchnic vasoconstriction (β2 blockade).

A. Metoprolol is β1-selective and lacks the splanchnic (β2) effect.

B. Octreotide is used in acute variceal bleeding.

79. D. No treatment; return if symptoms develop

Cholelithiasis · Step 2

Most gallstones never cause symptoms, and the risk of surgery outweighs the benefit in asymptomatic patients. Cholecystectomy is offered once symptoms (such as biliary colic) or complications occur.